SENIOR MATHS

CALCULUS

Kevin Dunne

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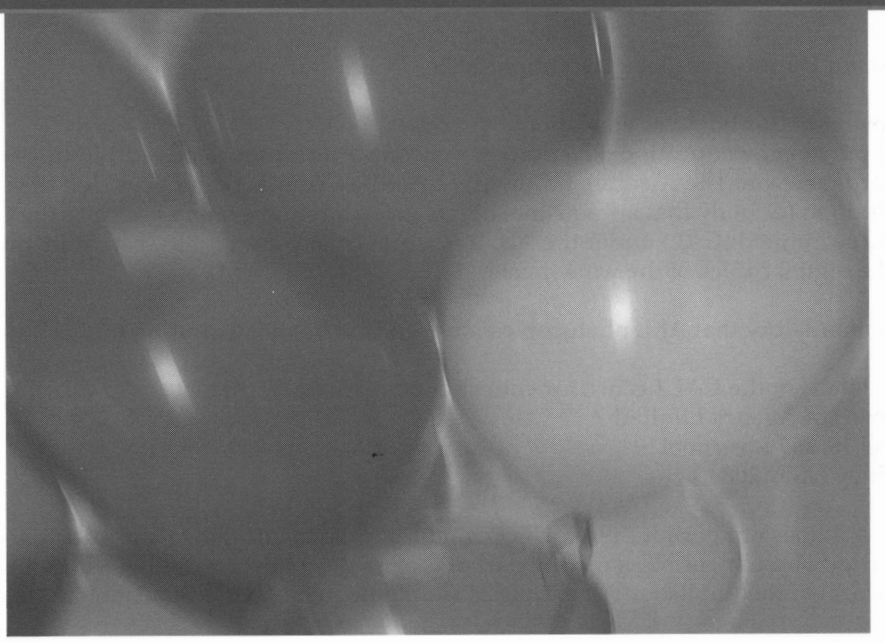
INSTANT LESSONS

SENIOR MATHS

CALCULUS



Kevin Dunne

INSTANT LESSONS

Kevin Dunne is a highly experienced secondary school maths teacher and a valued member of the Instant Lessons team of authors. He has taught for two years in primary schools and 32 years in secondary schools in New South Wales. He also taught for a year in New Zealand as an exchange teacher. Kevin has been a member of the NSW Senior Mathematics Syllabus Committee and the HSC Examination Committee. He marked HSC examination papers for 10 years. This is the fifth book he has written in this series of *Assignments in Senior Mathematics*.

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I would like to thank my wife, Rikki, for her invaluable assistance in the preparation of this book.

Kevin Dunne

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introduction

This series has been written to provide comprehensive coverage of selected topics from the senior mathematics syllabus. This book concentrates on Calculus. Each worksheet has been carefully graded with the more difficult questions towards the end of each set. Where appropriate, exercises include a worked example to show the most common methods of working. Students should be able to complete the worksheets with minimal help from the teacher. Each topic has sufficient questions to occupy students for at least a 40-minute lesson.

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There are several different methods for solving limits.

e.g.

1 Direct substitution:

$$\lim_{x\to 3}\frac{3x+4}{x+5}$$

$$=\frac{13}{8}$$

2 Where direct substitution results in a zero denominator, e.g. for $\lim_{x\to 2}\frac{x^2-4}{x-2}$, direct substitution results in 0/0, which is undefined, then we use the following procedure:

$$\lim_{x\to 2}\frac{x^2-4}{x-2}$$

$$=\lim_{x\to 2}\frac{(x+2)(x-2)}{(x-2)}$$

$$=\lim_{x\to 2}x+2$$

$$=4$$

3 For questions which use the fact that $\lim_{x\to\infty}\frac{1}{x}=0$ (or $\lim_{x\to\infty}\frac{1}{x^2}=0$ etc.) and cannot be solved by either of the above methods, use the following method:

$$\lim_{x\to\infty}\frac{4x^2+3}{x^2+1}$$ (divide each term by the highest power of x)

$$=\lim_{x\to\infty}\frac{\frac{4x^2}{x^2}+\frac{3}{x^2}}{\frac{x^2}{x^2}+\frac{1}{x^2}}$$

$$=\lim_{x\to\infty}\frac{4+\frac{3}{x^2}}{1+\frac{1}{x^2}}$$

$$=\frac{4+0}{1+0}$$

$$=4$$

---

1 $\lim_{x\to 4}5x$ **2** $\lim_{x\to 0}3x$ **3** $\lim_{x\to 2}4x+5$ **4** $\lim_{x\to 5}3x-5$

5 $\lim_{x\to -2}3x+4$ **6** $\lim_{x\to -4}x^2+4x$ **7** $\lim_{x\to 3}2x-x^2$ **8** $\lim_{x\to 5}2-x^2$

9 $\lim_{x\to 2}x^3+3x^2$ **10** $\lim_{x\to 4}\frac{x^2-16}{x-4}$ **11** $\lim_{x\to -4}\frac{x^2-16}{x+4}$ **12** $\lim_{x\to -1}\frac{x^2-1}{x+1}$

13 $\lim_{x\to 3}\frac{x^2-9}{x-3}$ **14** $\lim_{x\to 5}\frac{x^2-25}{x-5}$ **15** $\lim_{x\to -9}\frac{x^2-81}{x+9}$ **16** $\lim_{x\to 2}\frac{x^2+5x+1}{x+1}$

17 $\lim_{x\to 2}\frac{x^2-x-2}{x-2}$ **18** $\lim_{x\to -3}\frac{x^2+2x-3}{x+3}$ **19** $\lim_{x\to 5}\frac{x^2-2x-15}{x-5}$ **20** $\lim_{x\to 2}\frac{x^2-6x+8}{x-2}$

21 $\lim_{x\to\infty}\frac{x^2+3}{4x^2}$ **22** $\lim_{x\to\infty}\frac{3x+4}{2x-5}$ **23** $\lim_{x\to\infty}\frac{2x^2+x+1}{x^2-1}$ **24** $\lim_{x\to\infty}\frac{5x^2+6}{x^2-4x}$

25 $\lim_{x\to\infty}\frac{5x^3+4x^2}{6x^3+x}$ **26** $\lim_{x\to 4}\frac{x^2+4}{x+1}$ **27** $\lim_{x\to\infty}\frac{x^2-4}{x^2-2}$ **28** $\lim_{x\to 1}\frac{3x^2-3}{x-1}$

29 $\lim_{x\to 1}\frac{3x^2-3}{x+1}$ **30** $\lim_{x\to\infty}\frac{4}{x+2}$ **31** $\lim_{x\to\infty}\frac{5}{2+x}$ **32** $\lim_{x\to\infty}\frac{2x^3+3x^2+x}{x^3+4x^2+1}$

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When differentiating from first principles, the following formulae may be used.

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} \quad \text{or} \quad \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Note – Many symbols denote a derivative: $\frac{dy}{dx}$, y', $f'(x)$, $\frac{d}{dx} f(x)$

When we differentiate we find the **derivative** or the **gradient function** or the **differential coefficient**.

e.g.

1 Find from first principles the derivative of $y = x^2 + 3$.

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{(x+\Delta x)^2 + 3 - (x^2+3)}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{x^2 + 2x\Delta x + \Delta x^2 + 3 - x^2 - 3}{\Delta x}$$

$$= \lim_{\Delta x \to 0} \frac{2x\Delta x + \Delta x^2}{\Delta x}$$

$$= \lim_{\Delta x \to 0} 2x + \Delta x$$

$$= 2x$$

2 Find from first principles the derivative of $y = x^2 - 4x$.

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{(x+h)^2 - 4(x+h) - (x^2 - 4x)}{h}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - 4x - 4h - x^2 + 4x}{h}$$

$$= \lim_{h \to 0} \frac{2xh + h^2 - 4h}{h}$$

$$= \lim_{h \to 0} \frac{h(2x + h - 4)}{h}$$

$$= 2x - 4$$

Find the derivative from first principles:

1 $y = 2x$ **2** $y = 2x + 4$ **3** $y = 3x + 5$

4 $y = 4 - 5x$ **5** $y = \frac{1}{3}x + 4$ **6** $y = 2x^2$

7 $y = 3x^2$ **8** $y = 2x^2 + 5$ **9** $y = 6 - x^2$

10 $y = x^2 + x$ **11** $y = 4x^2 + 4x + 1$ **12** $y = 4x - x^2$

13 Complete the following example to determine from first principles the differential coefficient of x^3:

$$\frac{d}{dx}(x^3) = \lim_{h \to 0} \frac{f(x+h)^3 - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{(x+h)^3 - x^3}{h}$$

$$= \lim_{h \to 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h}$$

$$= \lim_{h \to 0} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

$$= \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

14 Find, from first principles, the derivative of $y = x^3 + 4$.

15 Differentiate $y = x^3 + 4x$ from first principles.

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If $f(x) = ax^n$ then $f'(x) = anx^{n-1}$. There are a number of ways of expressing a derivative. The examples following show some of these ways.

e.g.

1 Differentiate with respect to x.

i 6

$\frac{d}{dx}(6) = 0$

ii $-5x$

$\frac{d}{dx}(-5x) = -5$

iii $x^3 + 4x$

$\frac{d}{dx}(x^3 + 4x) = 3x^2 + 4$

2 Find the derivative of the following functions:

i $y = 5x^4 + 2x + 1$

$\frac{dy}{dx} = 20x^3 + 2$

ii $y = x^{-3} + x^3$

$\frac{dy}{dx} = -3x^{-4} + 3x^2$

iii $y = \frac{1}{x^2}$

$y = x^{-2}$

$\frac{dy}{dx} = -2x^{-3}$

3 Differentiate:

i $y = \frac{x^3 + 3x}{x^2}$

$y = \frac{x^3}{x^2} + \frac{3x}{x^2}$

$= x + \frac{3}{x}$

$y' = 1 - 3x^{-2}$

$= 1 - \frac{3}{x^2}$

ii $f(x) = (x+3)(x-5)$

$f(x) = x^2 - 2x - 15$

$f'(x) = 2x - 2$

4 Given $f(x) = 3x^3 - 5x$, evaluate $f'(3)$:

$f'(x) = 9x^2 - 5$

$f'(3) = 9 \times 3^2 - 5$

$= 76$

Differentiate with respect to x:

1 9 **2** 12 **3** $7x$

4 $6x$ **5** x^2 **6** $9x^2$

7 $-7x^2$ **8** x^4 **9** $-6x^4$

Find the derivative of:

10 $6x^3$ **11** $9x^{10}$ **12** $5x^3 + 4$ **13** $7x^2 + 6$

14 $8x^4 + 3x^2$ **15** $6x^2 - 7x + 1$ **16** $9x^3 + 8x - 5$ **17** $3x^4 + 6x^2 - x$

18 $3x^3 + 4x - 9$ **19** $7 - 4x$ **20** $9 - 14x^2$ **21** $3x - 7x^2$

Differentiate:

22 $y = 7x - 4$ **23** $y = 6x - 3x^2$ **24** $y = 8 - 3x^2 + 4x^3$ **25** $y = x^{-3}$

26 $y = 6x^{-2}$ **27** $y = 7x^{-3} + x^2$ **28** $y = \frac{1}{x}$ **29** $y = \frac{4}{x^2} + x$

30 $y = \frac{3}{x^2} + \frac{1}{x^3}$ **31** $f(x) = 4x(x+5)$ **32** $f(x) = 3(x^2 + 4x)$ **33** $y = (x+4)(x+3)$

34 $f(x) = (2x-1)(x+1)$ **35** $f(x) = (x+2)(3x-1)$ **36** $f(x) = (9x+1)(x^2+1)$ **37** $f(x) = \frac{x^2 + x}{x^2}$

38 $f(x) = \frac{x^2 + 3x}{x^3}$ **39** $f(x) = \frac{x^2 - 5x}{x^3}$ **40** $f(x) = \frac{x - 7x^5}{x^4}$ **41** $f(x) = \frac{x^3 + 3x^2 + x}{x^2}$

42 $f(x) = \frac{2x^3 - 3x - 1}{x}$

43 Given $f(x) = 6x^2 + 3x$, determine the value of **i)** $f'(3)$ **ii)** $f'(5)$.

44 Given $f(x) = 3x^3 + 4x$, determine the value of **i)** $f'(-1)$ **ii)** $f'(4)$.

45 Given $f(x) = 9x^2 + 4x + 6$, find the value of **i)** $f'(-5)$ **ii)** $f'(2)$.

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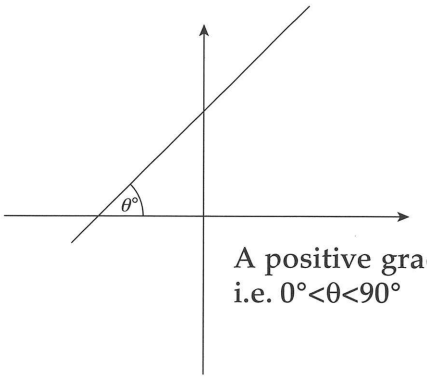
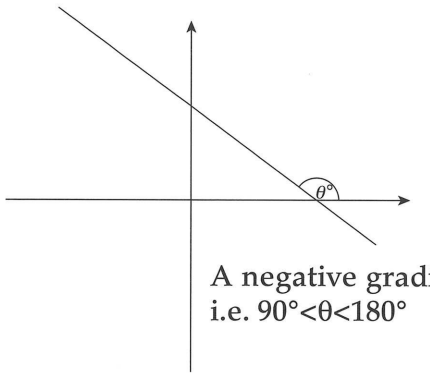
gradient function and angle of inclination

Differentiation produces a gradient function. From this function we can find the gradient of the curve at a given point. This gradient is also a measure of the slope of the tangent to the curve or the measure of the gradient of the tangent to the curve at the point (and consequently the "instantaneous rate of change"). It also represents the tangent of the angle of inclination.



A positive gradient,
i.e. $0°<\theta<90°$



A negative gradient,
i.e. $90°<\theta<180°$

e.g.

1 Determine the gradient of the function $y = 3x^2 + 2x$ at the point where $x = 2$.

$y = 3x^2 + 2x$

$\frac{dy}{dx} = 6x + 2$

At $x = 2$, $\frac{dy}{dx} = 14$

The gradient of the function is 14

2 For $f(x) = 2x^2 - 5x$, find $f'(x)$ and determine the angle of inclination of the tangent to the curve at $x = 1$.

$f(x) = 2x^2 - 5x$

$f'(x) = 4x - 5$

$f'(1) = -1$

$\tan\theta = -1$

$\theta = \tan^{-1}(-1)$

$\theta = 180° - 45°$

$= 135°$

Note: a negative gradient means θ is obtuse.

Determine the gradient functions of the following.

1 $7x + 4$ **2** $6x - 5$ **3** $4x + 8$ **4** $x^2 + 7x$

5 $x^3 - 3x$ **6** $5x - 9x^2$ **7** $3x(x + 1)$ **8** $4(5 - 3x^3)$

9 $(x + 4)(x + 3)$ **10** x^{-1} **11** $\frac{3}{x}$ **12** $\frac{6}{x^2}$

13 Determine the gradient of the curve $y = x^2 + 4x$ at the point where $x = 4$.

14 Determine the gradient of the curve $y = 2x^3 + 1$ at the point where $x = 1$.

15 Find the gradient of the curve $y = 2x^2 + 4$ at the point where $x = 4$.

16 Find the gradient of the curve $y = 3x^3 - 4x$ at the point where $x = -1$.

17 Find the gradient of the curve $y = x^3 - 2x$ at the point $(1, -1)$.

18 Find the gradient of the curve $y = x^3 - x^2$ at the point $(3, 18)$.

19 Find the gradient of the tangent to the curve $y = x^2 + 5x$ at the point where $x = 1$.

20 Find the gradient of the tangent to the curve $y = x^3 - 2x$ at the point $(2, 4)$.

21 Determine the angle of inclination of the tangent to $y = 6x^2 + 4x$ at the point where $x = 2$.

22 Determine the angle of inclination of the tangent to $y = 2x^5 + x^3$ at the point where $x = 1$.

23 Determine the angle of inclination of the tangent to $y = x^4 - 3x^2 - x$ at the point $(-1, -1)$.

24 Determine the angle of inclination of the tangent to $y = x^3 - x^2 - 3x$ at the point $(3, 9)$.

25 At a point the angle of inclination of a tangent to a curve is 60°. What is the gradient?

26 At a point the angle of inclination of a tangent to a curve is 135°. What is the gradient?

27 The angle of inclination of a tangent to the curve $y = x^2 - 5x + 1$ is 45°. Determine the coordinates of the point where the tangent touches the curve.

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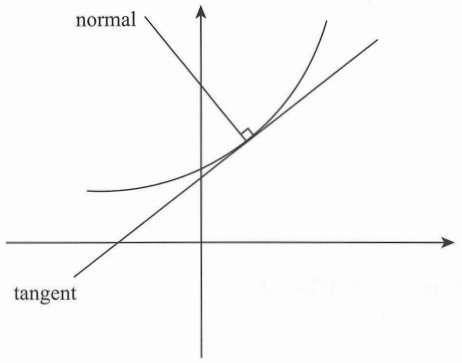
The equation of a tangent can be found by finding the gradient of the tangent at the point (using the derivative) and the coordinates of the point.

A **normal** is the line perpendicular to the tangent at the point of contact.



e.g. Find equations of the tangent and normal to the curve $y = 4x^2 - 6x + 3$ at the point $x = 2$.

$y' = 8x - 6$

At $x = 2$, $y = 4 \times 2^2 - 6 \times 2 + 3$

$= 7$

$y'(2) = 8 \times 2 - 6$

$= 10$

The gradient of the tangent is 10 and gradient of the normal is $-\frac{1}{10}$.

To find the equation of a straight line, use $m = \frac{y - y_1}{x - x_1}$

Equation of tangent:

$$10 = \frac{y-7}{x-2}$$

$$10x - 20 = y - 7$$

$$10x - y - 13 = 0$$

Equation of normal:

$$-\frac{1}{10} = \frac{y-7}{x-2}$$

$$x - 2 = -10y + 70$$

$$x + 10y - 72 = 0$$

Find the equations of the tangent and normal to the following curves:

1 $y = x^2 + 4x$ at the point $(1,5)$

2 $y = x^2 - 6x + 3$ at the point $(2,-5)$

3 $y = x^3 - 2x$ at the point $(-1,1)$

4 $y = \frac{3}{x}$ at the point $(3,1)$

5 $y = \frac{4}{x}$ at the point $(2,2)$

6 $y = x^4 + 1$ at the point where $x = 1$

7 $y = x^3 + 3x$ at the point where $x = 2$

8 $y = x^2 - 3x + 1$ at the point where $x = -3$

9 At what point on the curve $y = x^2 + 1$ does the gradient of the tangent equal 6? Determine the equation of the tangent at this point.

10 Determine the point on the curve $y = 3x^2 - 4$ where the gradient of the tangent is 12. Hence determine the equation of the tangent.

11 Determine the point on the curve $y = x^2 - 4x + 2$, where the gradient of the tangent is 2. Hence determine the equation of the normal, at that point.

12 Determine the point on the curve $y = x^4 - 5x$, where the gradient of the normal is 1. Hence determine the equation of the tangent, at that point.

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Find the derivative of the following functions from first principles:

1 $y=6x$ **2** $y=7x+4$ **3** $y=3-3x$

4 $y=5x^2$ **5** $y=3x^2+x$ **6** $y=7x^2-2x+1$

Find the following limits:

7 $\lim\limits_{x\to 3} 3x+4$ **8** $\lim\limits_{x\to 2} x^2+3x+1$ **9** $\lim\limits_{x\to 5} x^3-5x^2+6$

10 $\lim\limits_{x\to -2} \dfrac{x^2-4}{x+2}$ **11** $\lim\limits_{x\to 5} \dfrac{x^2-25}{x-5}$ **12** $\lim\limits_{x\to \infty} \dfrac{4x+5}{x-2}$

13 $\lim\limits_{x\to 8} \dfrac{x^2-64}{x-8}$ **14** $\lim\limits_{x\to 5} \dfrac{x^2-3x-10}{x-5}$ **15** $\lim\limits_{x\to -4} \dfrac{x^2+x-12}{x+4}$

16 $\lim\limits_{x\to \infty} \dfrac{3x^2+4x}{x^2+1}$ **17** $\lim\limits_{x\to \infty} \dfrac{2x^3+4x^2}{3x^3+7x+1}$ **18** $\lim\limits_{x\to \infty} \dfrac{x^2-9}{x^2-3}$

Find the derivative of the following functions:

19 6 **20** $4x^3$ **21** $-7x^5$

22 $5x-7x^2-x^4$ **23** $6-4x^3-x^6$ **24** x^{-4}

25 $-2x^2+x+x^{-3}$ **26** $4x^3-6x+4-x^{-1}$ **27** $x^5-\frac{1}{2}x^4+5-x^{-2}$

Find the gradient function of:

28 $7x-4$ **29** $6-8x+x^5$ **30** $2x-x^{-3}$

31 $4x^{-5}+1$ **32** $\dfrac{3}{x^2}$ **33** $\dfrac{x^2+x}{x^2}$

Differentiate:

34 $\dfrac{3}{x^3}+\dfrac{1}{x^2}$ **35** $1-\dfrac{4}{x^4}$ **36** $(x+3)(x-5)$

37 $3x(4x-6x^2)$ **38** $\dfrac{4x^3-3x^2+1}{x}$ **39** $\dfrac{x^4-x^3+x}{x^2}$

Determine the equations of the tangent and normal to the following curves:

40 $y=6x^3+7x+4$ at the point $(0,4)$

41 $y=2x-x^3$ at the point where $x=2$

42 $y=x^2+3x$ at the point where $x=-2$

43 $y=3x^4+x^2$ at the point where $x=1$

44 Determine the angle of inclination (to the nearest degree) of the tangent to the curve $y=4x^2-5x+1$ at the point where $x=1$.

45 Determine the angle of inclination (to the nearest degree) of the tangent to the curve $y=x^3-3x+3$ at the point where $x=2$.

46 Determine the angle of inclination (to the nearest degree) of the tangent to the curve $y=2x^3-5x-4$ at the point where $x=-1$.

47 The angle of inclination of the tangent at point A to the curve $y=x^2-3x+1$, is 45°. Determine the coordinates of point A.

48 The angle of inclination of the tangent at point B to the curve $y=2x^2+7x+3$ is 135°. Determine the coordinates of point B.

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Review of Index Laws:

$a^m \times a^n = a^{m+n}$ $\quad a^m \div a^n = a^{m-n}$ $\quad (a^m)^n = a^{mn}$

$a^m b^m = (ab)^m$ $\quad a^0 = 1$ $\quad a^{-n} = \dfrac{1}{a^n}$

$a^{\frac{1}{n}} = \sqrt[n]{a}$ $\quad a^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$

e.g.

1 Change to index form:

i $\dfrac{x^5}{x^3} = x^2$ **ii** $\sqrt[3]{x} = x^{\frac{1}{3}}$ **iii** $\sqrt[3]{x^2} = (x^2)^{\frac{1}{3}} = x^{\frac{2}{3}}$ **iv** $\dfrac{1}{\sqrt[3]{x^2}} = x^{-\frac{2}{3}}$

2 Differentiate with respect to x:

i $\dfrac{d}{dx}(x^{\frac{5}{3}}) = \dfrac{5}{3}x^{\frac{2}{3}}$

ii $y = \sqrt[4]{x}$

$= x^{\frac{1}{4}}$

$\dfrac{dy}{dx} = \dfrac{1}{4}x^{-\frac{3}{4}}$

Find the derivatives of the following functions.

1 $x^{\frac{1}{2}}$ **2** $x^{\frac{1}{3}}$ **3** $x^{\frac{1}{5}}$ **4** $x^{\frac{2}{3}}$

5 $x^{\frac{3}{2}}$ **6** $x^{\frac{3}{4}}$ **7** $2x^{\frac{1}{2}}$ **8** $3x^{\frac{3}{4}}$

9 $x^{-\frac{1}{3}}$ **10** $x^{-\frac{2}{3}}$ **11** $4x^{-\frac{3}{4}}$ **12** $4x^{-1\frac{1}{3}}$

Use the index laws to simplify the following, and then find their derivatives:

13 $x^3 \times x^5$ **14** $\dfrac{x^{14}}{x^7}$ **15** $x^3 \div x^4$ **16** $\dfrac{x^4}{x^6}$

17 $3\sqrt{x}$ **18** $4\sqrt[3]{x}$ **19** $\sqrt[5]{x}$ **20** $x\sqrt{x}$

21 $x^2\sqrt{x}$ **22** $x\sqrt[3]{x}$ **23** $x^3\sqrt{x}$ **24** $x^2\sqrt[3]{x}$

25 $\dfrac{x}{\sqrt{x}}$ **26** $x\sqrt[4]{x}$ **27** $\dfrac{x^2}{\sqrt{x}}$ **28** $\dfrac{x}{3\sqrt{x}}$

29 $\dfrac{x^2}{\sqrt[3]{x}}$ **30** $x^{\frac{2}{3}} \times x^3$ **31** $\dfrac{\sqrt{x}}{4}$ **32** $\dfrac{1}{\sqrt[5]{x}}$

33 $\dfrac{\sqrt{x}}{x}$ **34** $\dfrac{\sqrt[3]{x}}{5}$ **35** $\dfrac{x^{-\frac{2}{3}}}{3}$ **36** $\dfrac{1}{3x^{\frac{1}{3}}}$

37 $x + \sqrt{x}$ **38** $\sqrt{x} + 4x^2$ **39** $x^2 + x^{-\frac{1}{3}}$ **40** $\sqrt{x} - \dfrac{3}{\sqrt{x}}$

41 $\sqrt[5]{x} - \dfrac{1}{x}$ **42** $\sqrt[3]{x} - \dfrac{1}{\sqrt[3]{x}}$

43 Find the gradient of the tangent to the curve $y = \sqrt{x} + 3$ at the point $(4, 5)$.

44 Find the gradient of the tangent to the curve $y = \sqrt[3]{x} - 4$ at the point where $x = 1$.

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The derivative of a product is given by $\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$ where $y = u \times v$.

You may choose to remember this rule as:
'the first (factor) by the derivative of the second plus the second by the derivative of the first'.

e.g.

1 Differentiate $y = (x+3)(5x+2)$

$$\frac{dy}{dx} = (x+3) \times \frac{d}{dx}(5x+2) + (5x+2) \times \frac{d}{dx}(x+3)$$
$$= (x+3)5 + (5x+2)1$$
$$= 5x + 15 + 5x + 2$$
$$= 10x + 17$$

Note: We could have expanded first and then differentiated.

2 Differentiate $f(x) = (3x^2+4x+3)(x^2+7)$

$$f'(x) = (3x^2+4x+3) \times 2x + (x^2+7)(6x+4)$$
$$= 6x^3 + 8x^2 + 6x + 6x^3 + 4x^2 + 42x + 28$$
$$= 12x^3 + 12x^2 + 48x + 28 \text{ or}$$
$$= 4(3x^3 + 3x^2 + 12x + 7)$$

Use the product rule to differentiate:

1 $y = (x^2+2)(2x+4)$
2 $y = (x^2+3)(x-5)$
3 $y = (x^3+4)(2x-3)$
4 $y = (x^2-3x)(x+6)$
5 $y = (x^4+x)(2x+9)$
6 $y = (x^4+3x^2)(3x-9)$
7 $y = (x^2-4)(x^2-2x)$
8 $y = (2x^3+4x)(x^4-x)$
9 $y = (x^3-3x^2)(x^4+6x)$
10 $y = (x^3-4x)(2x^2-x)$

Differentiate:

11 $(2x+5)(x^3+4)$
12 $(7x^2+3x)(2x-3)$
13 $(x^3+6)(x^2+4)$
14 $(x^2+x)(x^3-x)$
15 $(x^2+3x+1)(x+6)$
16 $(x^3-3x)(2x^2+3x)$

Find $f'(x)$ for the following:

17 $f(x) = (6x+9)(4x^3-3x)$
18 $f(x) = (6x^3+8x)(x^2-1)$
19 $f(x) = (4x-3x^2)(2x^3+8x)$
20 $f(x) = (10x^2-2)(3x^3+4x^2)$
21 $f(x) = (2x^3+3x+1)(x-2)$
22 $f(x) = (3x^3-2x^2+1)(4x-3)$

23 If $f(x) = (x^2+2x)(x-1)$, find the derivative of $f(x)$ and hence the value of $f'(3)$.

24 Given $f(x) = (3x^2+4x)(x+3)$, find $f'(x)$ and hence the value of $f'(1)$.

25 If $f(x) = (x^2-3x)(x-x^2)$, find the value of $f'(0) + f'(1)$.

26 If $f(x) = (x-3x^2)(2-x^4)$, find the value of $f'(-1) + f'(1)$.

27 If $f(x) = (x^2+2x-1)(x-1)$, find the value of $f'(1) - f'(-1)$.

28 Find the gradient of the tangent to the curve $y = (3x+4)(x^2-2x)$ at the point where $x = 1$.

29 Find the gradient of the normal to the curve $y = (x^2+3x)(2x-3)$ at the point where $x = 2$.

30 Given $y = (x^3-3x)(x^2-2)$, find the equation of the tangent at the point where $x = 2$.

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If $y=\frac{u}{v}$ then the derivative is $\frac{dy}{dx}=\frac{v\frac{du}{dx}-u\frac{dv}{dx}}{v^2}$.

You may choose to remember the rule as 'bottom times the derivative of the top, minus the top times the derivative of the bottom, divided by the bottom squared'.

e.g. For $y=\frac{x^2-3}{2x-1}$, $\frac{dy}{dx}=\frac{(2x-1)\times 2x-(x^2-3)\times 2}{(2x-1)^2}$

$$=\frac{4x^2-2x-2x^2+6}{(2x-1)^2}$$

$$=\frac{2x^2-2x+6}{(2x-1)^2}$$

Find the derivatives of:

1 $y=\frac{x}{2x+1}$ **2** $y=\frac{-4x}{3x+2}$ **3** $y=\frac{x^2}{4x-5}$

4 $y=\frac{-x^3}{3x+4}$ **5** $y=\frac{x}{x^2+1}$ **6** $y=\frac{3x}{x^2-2}$

7 $y=\frac{-2x}{3x^2-x}$ **8** $y=\frac{3x+1}{2x+1}$ **9** $y=\frac{3-x}{4x-5}$

10 $y=\frac{x^2+1}{x-2}$ **11** $y=\frac{x^2-3}{x+1}$ **12** $y=\frac{3-x^2}{4-2x}$

Differentiate:

13 $\frac{x^2+1}{x^3}$ **14** $\frac{x^3}{x^2+1}$ **15** $\frac{4x^2}{6x+5}$

16 $\frac{2x^2}{x^2-1}$ **17** $\frac{x^3}{2x^2+1}$ **18** $\frac{-x}{x^3+1}$

19 Find the gradient of the tangent to the curve $y=\frac{2x+4}{x+1}$ at the point where $x=1$.

20 Find the gradient of the tangent to the curve $y=\frac{x-1}{x-2}$ at the point where $x=3$.

21 Determine the angle of inclination (to the nearest degree) to the positive x-axis, of the tangent to $y=\frac{x+1}{x+3}$ at the point where $x=1$.

22 Determine the angle of inclination (to the nearest degree) to the positive x-axis of the tangent to $y=\frac{x^2}{x-1}$ at the point where $x=-2$.

23 Find the equation of the tangent to the curve $y=\frac{x-3}{x+1}$ at the point $(-3,3)$.

24 Find the equation of the tangent to $y=\frac{3x-5}{3-4x}$ at the point where $x=1$.

25 Find the equation of the normal to the curve $y=\frac{x^3}{3x-2}$ at $x=-2$.

26 Given $f(x)=\frac{2x-1}{x+1}$, determine the equation of the tangent and normal to the curve at the point $\left(1,\frac{1}{2}\right)$.

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The Chain Rule for differentiating a function of a function is:

If $y = f(u)$ *and* $u = f(x)$ *then* $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

e.g.

1 Use the chain rule to find $\frac{dy}{dx}$ if $y = (x^3 - 4)^4$

Let $u = x^3 - 4$ thus $\frac{du}{dx} = 3x^2$ and $y = u^4$ thus $\frac{dy}{du} = 4u^3$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$\frac{dy}{dx} = 4u^3 \times 3x^2 \text{ but } u = x^3 - 4$$

$$\frac{dy}{dx} = 4(x^3 - 4)^3 \times 3x^2$$

$$= 12x^2(x^3 - 4)^3$$

Students are encouraged to perform this differentiation using the shorter method below:

2 Differentiate $\sqrt{4x-3}$

$$\frac{d}{dx}(4x-3)^{\frac{1}{2}} = \frac{1}{2}(4x-3)^{-\frac{1}{2}} \times 4$$

$$= 2(4x-3)^{-\frac{1}{2}}$$

$$= \frac{2}{\sqrt{4x-3}}$$

3 Find $f'(x)$ given $f(x) = (3x^2 - 4x)^5$.

$$f'(x) = 5(3x^2 - 4x)^4 \times (6x - 4)$$

$$= 5(6x - 4)(3x^2 - 4x)^4$$

Differentiate the following:

1 $(4x+2)^3$ **2** $(5x+3)^4$ **3** $(7-6x)^4$

4 $(7x-3)^5$ **5** $(x^2+4)^3$ **6** $(4x^2-7)^3$

7 $(2-3x^3)^4$ **8** $(3x^4+6)^5$ **9** $(7x^2-3)^{-1}$

10 $(3x-1)^{-1}$ **11** $\frac{1}{3-4x^2}$ **12** $\frac{1}{2x^2+1}$

13 $\sqrt{4x+2}$ **14** $\sqrt{3x-5}$ **15** $\sqrt{3-2x^2}$

16 $(7x-3)^{-2}$ **17** $(6-x)^{\frac{1}{3}}$ **18** $\sqrt[3]{(2x^2-1)}$

19 $4\sqrt{6x-7}$ **20** $3\sqrt{6-2x}$ **21** $\frac{3}{\sqrt{2-4x}}$

22 Given $f(x) = (3-x^2)^3$, find the value of $f'(2)$.

23 Given $f(x) = (4-3x^2)^4$, find the value of $f'(3)$.

24 Given $f(x) = \sqrt{x^2-1}$, find the value of $f'(4)$.

25 Determine the gradient of the tangent to the curve $y = (3x-2)^3$ at the point where $x = 1$.

26 Find the gradient of the tangent and the normal to the curve $y = (x^2+1)^4$ at the point $(1,16)$.

27 Find the equation of the tangent and the normal to the curve $y = (2-x^2)^3$ at the point $(1,1)$.

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e.g.

1 $y=\sqrt{x}$

$y=x^{\frac{1}{2}}$

$\frac{dy}{dx}=\frac{1}{2}x^{-\frac{1}{2}}$

$=\frac{1}{2\sqrt{x}}$

2 $f(x)=\frac{1}{\sqrt[3]{x}}$

$f(x)=x^{-\frac{1}{3}}$

$f'(x)=-\frac{1}{3}x^{-\frac{4}{3}}$

$=\frac{-1}{3\sqrt[3]{x^4}}$

3 $y=(x^2+3)(x^3+1)$

$y'=(x^2+3)3x^2+(x^3+1)2x$

$=3x^4+9x^2+2x^4+2x$

$=5x^4+9x^2+2x$

4 $f(x)=(2x-x^2)^3$

$f'(x)=3(2x-x^2)^2(2-2x)$

$=3(2-2x)(2x-x^2)^2$

5 $y=\frac{(2x+1)^3}{x}$

$y'=\frac{x\times 3(2x+1)^2\times 2-(2x+1)^3\times 1}{x^2}$

$=\frac{(2x+1)^2(6x-2x-1)}{x^2}$

$=\frac{(2x+1)^2(4x-1)}{x^2}$

Differentiate:

1 $x^{\frac{2}{3}}$

2 $(x+3)(x^2-2)$

3 $\frac{2x}{x+1}$

4 $(3+2x)^3$

5 $\frac{1}{x+3}$

6 $7\sqrt{x}$

7 $(x^2-3)(x^2+4)$

8 $\frac{3x}{x-5}$

9 $(3-4x)^4$

10 $\frac{1}{x^2-3}$

11 $(3x^2-1)(2x^3+2)$

12 $x^2\sqrt{x}$

13 $\frac{4x}{x^2+3}$

14 $(3-4x^2)^2$

15 $\frac{1}{3x-x^2}$

Find $\frac{dy}{dx}$ for the following functions:

16 $y=x\sqrt[3]{x}$

17 $y=(x^2-x)(x^3+1)$

18 $y=\frac{x^2}{2-3x^2}$

19 $y=(x^2+3)^5(3-x^3)$

20 $y=\frac{2}{x^3+3x}$

21 $y=\frac{x^3}{4\sqrt{x}}$

22 $y=(x^3+3x+4)^4$

23 $y=\frac{2}{x^3-4x}$

24 $y=(x^2+3x)(x^3-1)^2$

25 $y=\frac{x}{(x-1)^2}$

26 $y=\frac{2x}{(x+1)^3}$

27 Find the gradient of the tangent and normal to the curve $y=\frac{3}{\sqrt{x}}$ at the point where $x=9$.

28 Find the gradient of the tangent and normal to the curve $y=\frac{1}{x^2+1}$ at the point $\left(2,\frac{1}{5}\right)$.

29 Determine the equation of the tangent to the curve $y=(x^2+1)(2x-3)$ at the point $x=1$.

30 Given $f(x)=(x^2-4)^3$, determine $f'(1)-f'(-1)$.

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A function $y = f(x)$ is:

- increasing when $f'(x) > 0$
- decreasing when $f'(x) < 0$
- neither increasing nor decreasing, i.e. stationary, when $f'(x) = 0$.

e.g.

1 Determine whether the function $y = x^3 + 3x^2 + 4x$ is increasing, decreasing or stationary at the point $(0,0)$.

$y = x^3 + 3x^2 + 4x$

$y' = 3x^2 + 6x + 4$

When $x = 0$, $y' = 3 \times 0 + 6 \times 0 + 4$

$= 4.$

As $y' > 0$, the function is increasing at the point (0,0).

2 Find the range of values for which $y = \frac{1}{3}x^3 + \frac{3}{2}x^2 + 2x + 4$ is **i)** increasing **ii)** decreasing.

$\frac{dy}{dx} = x^2 + 3x + 2$

i) For an increasing function, $\frac{dy}{dx} > 0$, i.e.

$x^2 + 3x + 2 > 0$

$(x+2)(x+1) > 0$

$x < -2$ or $x > -1$

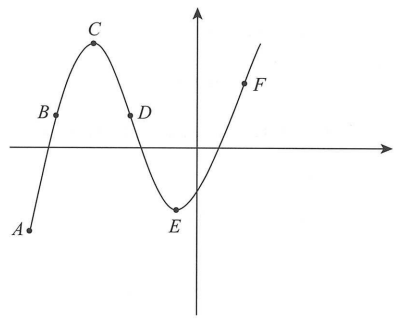
ii) For a decreasing function, $\frac{dy}{dx} < 0$, i.e.

$x^2 + 3x + 2 < 0$

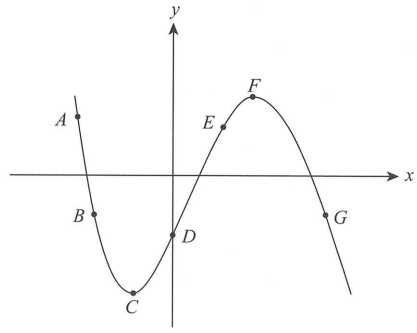
$-2 < x < -1.$

---

1 Write down the sign of the derivative and whether the function is increasing, decreasing or stationary at the points on the graph.



2 Write down the sign of the derivative at each of the points on the graph.



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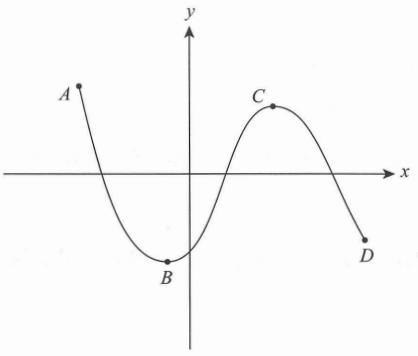
3 Between which two letters is the curve:

i) increasing?

ii) decreasing?



4 Show that the curve $y = x^3 - \frac{3}{2}x^2 - 3x + 2$ is:

i) increasing at the point where

a) $x = -3$ **b)** $x = -2$

ii) decreasing at the point where

a) $x = 0$ **b)** $x = 1$

5 For the curve $y = x^2 + 4x + 3$, determine whether the function is increasing, decreasing or stationary at the points where:

i) $x = -2$

ii) $x = 0$

iii) $x = 3$

iv) $x = -5$.

6 Show that $y = x^3$ is increasing for all values of x except where $x = 0$.

7 Determine where the function $y = x^2 - 2x - 3$ is increasing.

8 Show that $y = x^2 + x - 6$ is decreasing where $x < -\frac{1}{2}$.

9 Determine when $y = x^2 - 8x + 15$ is decreasing.

10 Find the x-values for which $y = x^3 - 3x^2 - 9x + 4$ is:

i) stationary

ii) decreasing.

11 Determine the values of x for which $y = -x^3 + 4x^2 + 3x - 1$ is:

i) stationary

ii) decreasing.

12 For what values of x is the function $y = -\frac{2}{3}x^3 + \frac{11}{2}x^2 - 12x + 4$:

i) increasing?

ii) decreasing?

13 Determine when $y = x^3 - 3x^2 - 24x + 3$ is stationary.

14 Determine when $y = -2x^3 + 11x^2 + 8x + 4$ is:

i) stationary

ii) increasing

iii) decreasing.

15 Determine whether the curve is increasing or decreasing at the points where $y = x^2 + 5x + 4$ cuts the x-axis and the y-axis.

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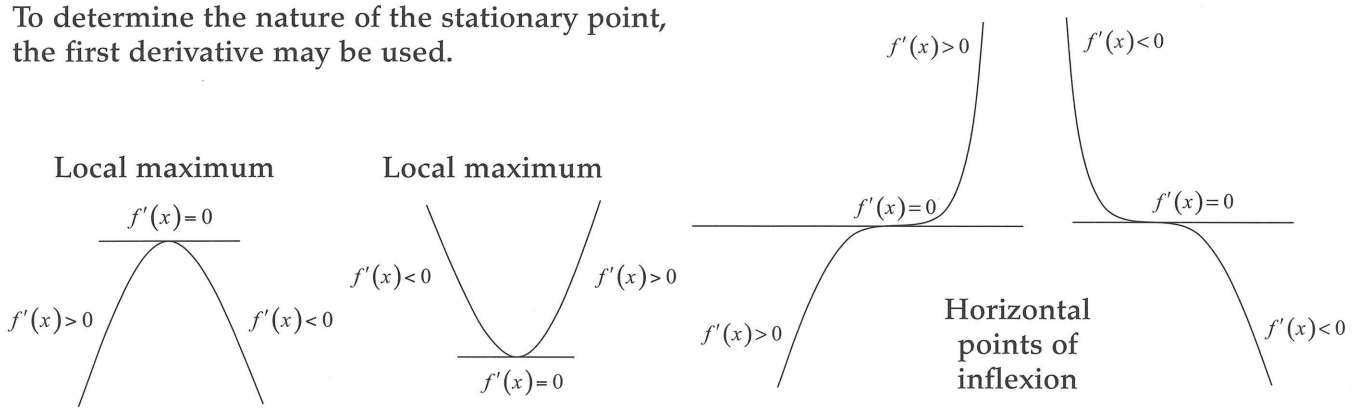
local (relative) maxima and minima and horizontal points of inflexion using the first derivative

13

A **stationary point** occurs when the tangent to the curve at that point is parallel to the x-axis (i.e. has a gradient of zero). A stationary point can be a **turning point** or a **horizontal point of inflexion**. At a point of inflexion, the tangent cuts the curve. Not all inflexion points are horizontal.

Turning points are local maxima or minima.

To determine the nature of the stationary point, the first derivative may be used.



e.g. Find the stationary point on the curve $y = x^2 + 3x + 4$ and determine its nature.

$y = x^2 + 3x + 4$

$y' = 2x + 3$

For stationary points, let $y' = 0$

Thus $2x + 3 = 0$

$x = -\frac{3}{2}$

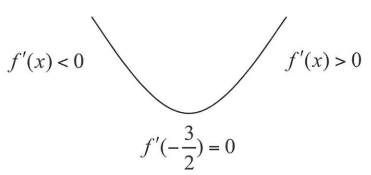
If $x = -\frac{3}{2}$

$y = \left(-\frac{3}{2}\right)^2 + 3 \times -\frac{3}{2} + 4$

$= 1\frac{3}{4}$

To determine the nature of the stationary point, a table is helpful.

x	-2	$-\frac{3}{2}$	-1
$f'(x)$	<0	0	>0



Thus a local minimum exists at the point $\left(-1\frac{1}{2}, 1\frac{3}{4}\right)$

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Given the following tables, state the nature of each of the six different stationary points at $x=3$.

1

x	$2\frac{1}{2}$	3	$3\frac{1}{2}$
$f'(x)$	>0	0	<0

2

x	2	3	4
$f'(x)$	>0	0	>0

3

x	2	3	4
$f'(x)$	>0	0	<0

4

x	$2\frac{1}{2}$	3	$3\frac{1}{2}$
$f'(x)$	<0	0	<0

5

x	2.25	3	3.1
$f'(x)$	<0	0	>0

6

x	2.9	3	3.1
$f'(x)$	<0	0	<0

Given the following information, state the nature of the stationary point at the point where $x = a$.

7 $f'(a)=0$

$f'\left(a+\frac{1}{4}\right)>0$

$f'\left(a-\frac{1}{4}\right)>0$

8 $f'(a)=0$

$f'\left(a+\frac{1}{4}\right)>0$

$f'\left(a-\frac{1}{4}\right)<0$

9 $f'(a)=0$

$f'(a+1)>0$

$f'(a-1)<0$

10 $f'(a)=0$

$f'\left(a+\frac{1}{2}\right)<0$

$f'\left(a-\frac{1}{4}\right)<0$

11 Show that $y=x^2+4x+1$ has a minimum turning point at $(-2,-3)$.

12 Show that $y=-x^2+6x-1$ has a maximum turning point at $(3,8)$.

13 Show that $y=x^3+4$ has a horizontal point of inflexion at the point $(0,4)$.

14 Show that $y=x^4+4$ has a minimum turning point at $(0,4)$.

15 Determine the coordinates of the stationary point of the curve $y=-x^4-6$.

16 Determine the coordinates of any stationary points of $y=x^2+8x+5$ and state their nature.

17 Find the coordinates and the nature of any turning points for $y=-4x^2+4x+3$.

18 Determine the coordinates of the stationary points of $y=2x^3-6x$ and state their nature.

19 Show that the curve $y=2x^3+15x^2+24x+12$ has 2 stationary points. Find the coordinates of the points and determine their nature.

20 The curve $y=x^3+3x^2+3x+1$ has one stationary point. Determine its nature.

21 Show that $y=(2x-3)^3$ has a horizontal point of inflexion. Determine its coordinates.

22 Determine the coordinates of the stationary points on the curve $y=\frac{1}{3}x^3-x^2-3x+4$ and determine their nature.

Determine the nature of the stationary points on each of these curves.

23 $y=-x^2+4x+5$

24 $y=x^2+2x-6$

25 $y=x^3-3x$

26 $y=2x^3+9x^2+48$

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The second derivative of a function is obtained by finding the derivative of the first derivative. There are several notations for the second derivative:

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) \qquad \frac{d}{dx}(f'(x)) \qquad \frac{d^2y}{dx^2} \qquad f''(x) \qquad y''$$

e.g. Find the second derivative of:

1 x^3+4x+1

$\frac{d}{dx}(x^3+4x+1)=3x^2+4$

$\frac{d}{dx^2}(3x^2+4)=6x.$

2 $y=x^2-5x+3$

$\frac{dy}{dx}=2x-5$

$\frac{d^2y}{dx^2}=2$

3 $f(x)=(x+4)^3$

$f'(x)=3(x+4)^2$

$f''(x)=6(x+4).$

4 Given $f(x)=4x^3-6x+5$, evaluate

$f(1)+f'(2)+f''(3)=$

$f'(x)=12x^2-6$

$f''(x)=24x$

$f(1)+f'(2)+f''(3)=(4-6+5)+(48-6)+72$

$=117$

Find the second derivative of the following:

1 $y=4x^2+5x+6$ **2** $y=6x^2+8x+5$ **3** $y=4x^3+3x$

4 $y=3x^3+7x-5$ **5** $y=x^2+x^{-1}$ **6** $y=\frac{1}{x^2}$

7 $y=6-5x-4x^3$ **8** $y=7-4x-5x^3$ **9** $y=x^5-3x^4+x-5$

10 $y=x-7x^3+5x^4-x^6$

Determine the second derivative of:

11 $f(x)=\frac{3}{2x}$ **12** $f(x)=\frac{x^2+1}{x}$ **13** $f(x)=\sqrt{x}$

14 $f(x)=\sqrt[3]{x}$ **15** $f(x)=(8x+1)^3$ **16** $f(x)=(3-4x)^5$

17 $f(x)=\sqrt{3x+1}$ **18** $f(x)=\sqrt{3-x}$ **19** $f(x)=\frac{x}{x-1}$

20 $f(x)=\frac{3x}{2-x}$

21 Given $f(x)=2x^3-5x^2+1$, find

i) $f''(3)$ **ii)** $f''(0)$ **iii)** $f''(1)+f'(1).$

22 Given $f(x)=3x+5x^2-x^3$, find

i) $f''(2)$ **ii)** $f''(-1)$ **iii)** $f''(-2)+f'(1)+f(0).$

23 If $f(x)=(x-1)^3$, find the values of

i) $f''(3)$ **ii)** $f''(-2)$ **iii)** $f'(3)-f(4).$

24 If $y=x+\frac{1}{x}$, show that

i) $y+\frac{d^2y}{dx^2}=\frac{x^4+x^2+2}{x^3}$ **ii)** $\frac{dy}{dx}+\frac{d^2y}{dx^2}=\frac{x^3-x+2}{x^3}.$

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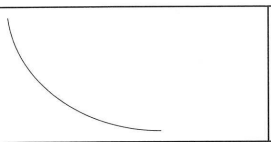
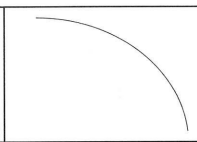
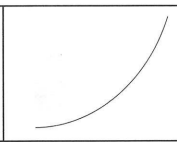
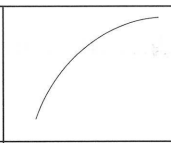
The sign of the second derivative can be used to determine the concavity of a function.
The second derivative in conjunction with the sign of the first derivative gives an indication of the slope of the curve or the rate of change of the function.
The following table summarises the significance of the sign of the first and second derivative.

			
Decreasing Concave up $y' < 0$ $y'' > 0$	Decreasing Concave down $y' < 0$ $y'' < 0$	Increasing Concave up $y' > 0$ $y'' > 0$	Increasing Concave down $y' > 0$ $y'' < 0$

e.g. Determine the concavity of the curve $y = x^3 + 3x + 4$ at points where:

i) $x = 3$ **ii)** $x = -1$

$y = x^3 + 3x + 4$

$\frac{dy}{dx} = 3x^2 + 3$

$\frac{d^2y}{dx^2} = 6x$

i) When $x = 3$ $\frac{d^2y}{dx^2} = 18 > 0$ Thus concave up

ii) When $x = -1$ $\frac{d^2y}{dx^2} = -3 < 0$ Thus concave down

---

Sketch a section of a graph, which satisfies the following conditions:

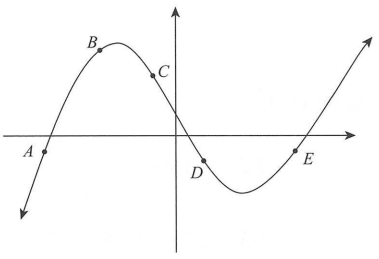
1 $f'(a) > 0$ and $f''(a) < 0$

2 $f'(a) < 0$ and $f''(a) < 0$

3 $f'(a) > 0$ and $f''(a) > 0$

4 $f'(a) < 0$ and $f''(a) > 0$

5 From this graph, write down the sign of the first and second derivatives at the points indicated:



By determining the first and second derivatives, describe the slope of the curve at the point indicated.

6 $y = x^2$ at $(-1,1)$

7 $y = x^2$ at $(2,8)$

8 $y = x^2 - 3x + 4$ at $(2,2)$

9 $y = -2x^2 + 4x - 8$ at $(2,-8)$

10 $y = x^3 + 2x^2 - 4x + 5$ at $(-3,8)$

11 $y = 3 - 2x - x^2$ at $(0,3)$

12 $y = x^3 - 4x^2 + 3x + 1$ at $(2,-1)$

13 $y = x^4 - 3x^3$ at $(2,-8)$

14 Determine the domain for which $y = x^2 + 3x + 4$ is concave up.

15 Show that the curve $y = 3 - 2x - x^2$ is concave down.

16 Determine the domain for which $y = \frac{1}{3}x^3 - 2x^2 + 3x + 5$ is concave down.

17 Determine the range of values for x, for which the curve $y = \frac{1}{3}x^3 + x^2 - 8x + 1$ is concave up and increasing.

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maxima, minima and points of inflexion using the second derivative

16

Maxima, minima and points of inflexion can be determined by using the second derivative.
The following tests are used:

Max. turning point	Min. turning point	Horizontal inflexion point	Inflexion point
		$f'(a)=0$	—
$f'(a)=0$ $f''(a)<0$	$f'(a)=0$ $f''(a)>0$	$f''(a)=0$ $f''(a^+)$ and $f''(a^-)$ have different signs	$f''(a)=0$ $f''(a^+)$ and $f''(a^-)$ have different signs

Note: Maximum and minimum turning points and horizontal points of inflexion are always stationary points.
At an inflexion point, the tangent cuts the curve.

e.g. For the function $y=\frac{1}{3}x^3+\frac{1}{2}x^2-12x+2$, find the stationary points, determine their nature and find the point of inflexion.

$y=\frac{1}{3}x^3+\frac{1}{2}x^2-12x+2$

$\frac{dy}{dx}=x^2+x-12$

$\frac{d^2y}{dx^2}=2x+1.$

For stationary points, let $\frac{dy}{dx}=0$

i.e. $x^2+x-12=0$

$(x+4)(x-3)=0$

Stationary points occur at $x = 3$, $x = -4$

When $x=3$, $y=-20\frac{1}{2}$ and $\frac{d^2y}{dx^2}>0$ and when $x=-4$, $y=36\frac{2}{3}$, $\frac{d^2y}{dx^2}<0$

Minimum turning point at $\left(3,-20\frac{1}{2}\right)$ and maximum turning point at $\left(-4,36\frac{2}{3}\right)$

For a point of inflexion, let $\frac{d^2y}{dx^2}=0$, i.e. $2x+1=0$

$x=-\frac{1}{2}$

Test points either side of $x=-\frac{1}{2}$, say at $x=-1$, $\frac{d^2y}{dx^2}<0$ and $x=0$, $\frac{d^2y}{dx^2}>0$, so concavity changes

Point of inflexion at $\left(-1,14\frac{1}{6}\right)$

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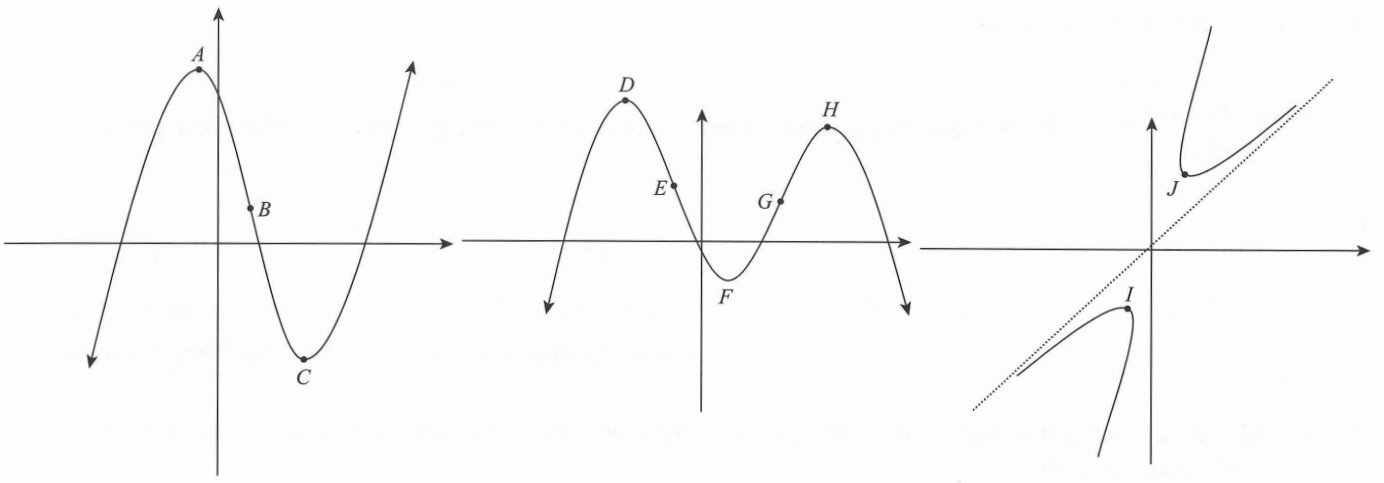

For the graphs sketched above, write down the points that satisfy the following conditions:

i) $\frac{dy}{dx}=0$, $\frac{d^2y}{dx^2}>0$

ii) $\frac{dy}{dx}=0$, $\frac{d^2y}{dx^2}<0$

iii) $\frac{d^2y}{dx^2}=0$, and $\frac{d^2y}{dx^2}$ changes signs either side of the point.

For each of the following curves, determine the point of inflexion. (Remember, test either side of the point.)

2 $y=\frac{1}{3}x^3-x^2+x-5$

3 $y=3x^3-6x^2+9x+15$

4 $y=x^4-24x^2+12x+1$

5 $y=-x^3+\frac{3}{2}x^2+x-5$

Show that these functions have only one stationary point and no points of inflexion.

6 $y=x^4-32x+4$

7 $y=-5+108x-x^4$

8 Show that $y=x^3+2x^2-4x+5$ has stationary points at $\left(\frac{2}{3},3\frac{14}{27}\right)$ and $(-2,13)$. Determine their nature, and find the x-coordinates of the point of inflexion.

9 Show that $y=4x+\frac{2}{x^2}$ has a minimum turning point at $(1,6)$. Show that $\frac{d^2y}{dx^2}=\frac{12}{x^4}$, and hence $\frac{d^2y}{dx^2}\neq 0$. Comment on the significance of this on any points of inflexion.

For the following functions, find the coordinates of the stationary points and determine the nature of these stationary points. Also find the points of inflexion, if they exist.

10 $y=x^2+8x-2$

11 $y=3-4x-x^2$

12 $y=x^4-6$

13 $y=4-x^4$

14 $y=x^3-7$

15 $y=x^3-12x+5$

16 $y=x^3-3x^2-9x+3$

17 $y=2x^3-15x^2$

18 $y=x^4-4x$

19 $y=-x^3+\frac{3}{2}x^2+9$

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e.g. For the function $y = x^3 - 4x^2 + 4x + 1$:

i) find the stationary points and any points of inflexion

ii) find the y-intercept

iii) sketch the curve.

i) $y = x^3 - 4x^2 + 4x + 1$

$\frac{dy}{dx} = 3x^2 - 8x + 4$

$\frac{d^2y}{dx^2} = 6x - 8$

For stationary points, let $\frac{dy}{dx} = 0$

$3x^2 - 8x + 4 = 0$

$(3x - 2)(x - 2) = 0$

$x = \frac{2}{3}$ or $x = 2$

When $x = \frac{2}{3}$, $y = 2\frac{5}{27}$ and $y'' = -4$

Maximum turning point at $\left(\frac{2}{3}, 2\frac{5}{27}\right)$.

When $x = 2$, $y = 1$ and $y'' = 4$

Minimum turning point at $(2,1)$

For point(s) of inflexion, let $\frac{d^2y}{dx^2} = 0$

$6x - 8 = 0$

$x = 1\frac{1}{3}$

When $x = 1\frac{1}{3}$, $y = 1\frac{16}{27}$

Test: when $x = 1$, $\frac{d^2y}{dx^2} < 0$

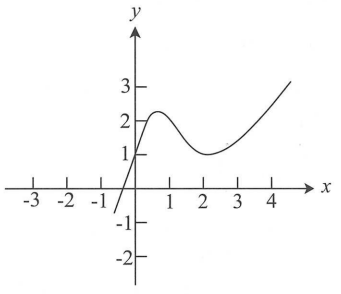
When $x = 2$, $\frac{d^2y}{dx^2} > 0$

change of sign, thus a point of inflexion at $\left(1\frac{1}{3}, 1\frac{16}{27}\right)$.

ii) For the y-intercept, let $x = 0$, thus $y = 1$

Note: When sketching a function, it is often helpful to determine what happens when x becomes very large and very small. i.e. in the above example, as $x \to \infty, y \to \infty$ and as $x \to -\infty, y \to -\infty$.

iii) To sketch the curve, we have a maximum turning point at $\left(\frac{2}{3}, 2\frac{5}{27}\right)$, a minimum turning point at $(2,1)$, a point of inflexion at $\left(1\frac{1}{3}, 1\frac{16}{27}\right)$, y-intercept at 1 as $x \to \infty, y \to \infty$ and $x \to -\infty, y \to -\infty$



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1 Show that the curve $y = x^3 - \frac{3}{2}x^2 + 4$ has stationary points at $x = 0$ and $x = 1$. Determine their nature. Find the coordinates of the point of inflexion and the y-intercept and hence sketch the function.

2 Show that $y = x^4 - 2x^2 + 5$ has stationary points at $x = 0$ and $x = \pm 1$. Determine their nature.

Show that the function also has two inflexion points at $x = \pm\frac{1}{\sqrt{3}}$. Sketch the function.

3 Show that the graph of $y = x^3 + 4$ has a horizontal point of inflexion and no turning points. Sketch the curve.

Draw sketches of these functions, showing any turning points, points of inflexion and intercepts (where they are easy to determine).

4 $y = x^2 + 6x - 7$

5 $y = x^2 - 6x + 5$

6 $y = 4 - 3x - 3x^2$

7 $y = x^3 - 3x^2 + 3x + 5$

8 $y = x^3 - 3x^2 + 1$

9 $y = -x^3 + 6x^2 - 9x + 1$

10 $y = -x^3 + 6x^2 + 4$

11 $y = x^4 - 18x^2 + 28$

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1 Show that $y = x^2 + 4x + 8$ is decreasing at the point $(-3, 5)$. For what values of x is the function increasing?

2 Show that $y = -x^3 + 4$ is decreasing for all values of x except $x = 0$.

3 Determine the values for which the graph of the function $2x^3 + 3x^2 - 72x + 6$ is increasing.

4 For the curve $y = -2x^3 + 3x^2 + 12x - 1$, complete the following tables and hence write down the nature of the stationary point at the points where $x = 2$ and $x = -1$.

i)

x	1	2	3
$f'(x)$			

ii)

x	-2	-1	0
$f'(x)$			

5 A curve $y = f(x)$ has a horizontal point of inflexion at $x = 2$. Which of the following could be true?

i) $f'(2) = 0$, $f'(2.1) > 0$, $f'(1.9) < 0$

ii) $f'(2) = -1$, $f'(2.1) < 0$, $f'(1.9) > 0$

iii) $f'(2) = 0$, $f'(2.1) > 0$, $f'(1.9) > 0$

iv) $f'(2) = 0$, $f'(2.1) < 0$, $f'(1.9) < 0$

Find the second derivative of the following functions.

6 $y = x^2 + 4x$

7 $y = 6 - x^2 + 5x$

8 $y = 4x^2 - 6x^3 + 5x^4$

9 $y = x^4 + \frac{1}{3}x^3 - \frac{1}{4}x^2$

10 $y = (4x - 3)^4$

11 $y = \sqrt{2x - 1}$

12 $y = \frac{1}{2x + 1}$

13 $y = (2x^2 + 1)(x^3 + 1)$

14 $y = (x^3 + x)(2x^4 - 1)$

15 Determine the concavity of the curve $y = x^3 + 4x^2 - 6x + 1$ at the point where $x = -3$.

16 Determine the range of values for which $y = -x^3 + 6x^2 + 4x + 1$ is

i) concave up **ii)** concave down.

17 Determine the coordinates of the point on the curve $y = x^3 - 9x^2 + 6x + 2$ where the concavity changes.

18 Show that $y = x^4 + 4x + 1$ has one stationary point. Determine its nature.

19 Draw a neat sketch of $y = x^3 + \frac{3}{2}x^2 - 18x + 3$ by first determining the stationary points and the x-coordinate of the point of inflexion.

20 A company reported that over time (T), its profits (P) continue to increase but that the profit growth rate was slowing down. Given the report, write down the signs of $\frac{dP}{dT}$ and $\frac{d^2P}{dT^2}$.

21 The attendance at the football over a number of years is given in the table below.

Year	1990	1991	1992	1993
Attendance	1000	1200	1500	2000

What is the sign of $\frac{dN}{dt}$ and $\frac{d^2N}{dt^2}$, where N represents the number of people and t represents time?

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e.g. Determine the maximum and minimum values of the function $2x^3 - 3x^2 - 12x$ in the range $-1 \le x \le 4$.

Note: To solve this type of question, it is helpful to:

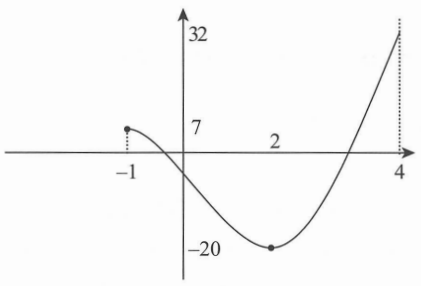
i) determine the maximum and minimum stationary points
ii) determne the coordinates of the points at the extremities of the domain
iii) sketch the curve.

$y = 2x^3 - 3x^2 - 12x$
$y' = 6x^2 - 6x - 12$
$y'' = 12x - 6$

For stationary points, let $y' = 0$, i.e. $6x^2 - 6x - 12 = 0$

$$(x-2)(x+1) = 0$$
$$x = 2 \text{ or } x = -1$$

At $x = 2$, $y = -20$, and $y'' = 18 > 0$ Thus minimum turning point
At $x = -1$, $y = 7$ and $y'' = -18 < 0$ Thus maximum turning point



Domain extremity values

At the point where $x = -1$, $y = 7$ but at $x = 4$, $y = 32$

Thus the maximum value is 32 and the minimum value is –20.

Determine maximum and minimum values of each of the following functions in the given domain:

1 $y = x^2 + 2x + 5, -2 \le x \le 3$

2 $y = x^2 - 4x - 5, 0 \le x \le 6$

3 $y = x^2 - 6x + 8, -2 \le x \le 5$

4 $y = -4 - 3x - x^2, -5 \le x \le 3$

5 $y = 3 + 4x - 2x^2, -4 \le x \le 2$

6 $y = 2x^3 - 3x^2 + 24, -1 \le x \le 1$

7 $y = -x^3 + 6x^2 - 9x + 1, 0 \le x \le 5$

8 $y = -2x^3 + 9x^2 + 24x - 1, -2 \le x \le 2$

9 $y = x^3 + 3x^2 + 3x + 1, 0 \le x \le 4$

10 $y = \frac{1}{3}x^3 - \frac{1}{2}x^2 + 4, -1 \le x \le 2$

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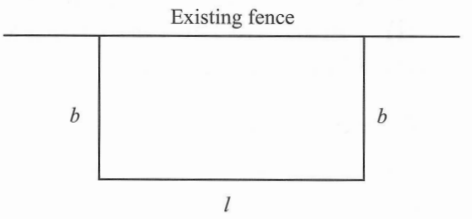
Calculus may be used to solve maxima and minima problems.

e.g. A preschool owner wants to make a secure play area for his preschoolers. He has 60 metres of fencing. If he uses an existing side fence, what is the maximum area he can enclose?



Let l be the length of the play area and b the breadth.

Then: $A = l \times b$ **(1)**

From the diagram: $l + 2b = 60$ **(2)**

From **(2)**: $l = 60 - 2b$

Substitute into **(1)**: $A = (60b - 2b) \times b$

$A = 60b - 2b^2$ **(3)**

Now we apply the process for finding maxima/minima.

$\frac{dA}{db} = 60 - 4b$ $\quad \frac{d^2A}{db^2} = -4$ which is < 0 Area is a maximum.

Let $\frac{dA}{db} = 0$ i.e. $60 - 4b = 0$

$b = 15$ Maximum area occurs when $b = 15$.

To find the area, we substitute into equation **(3)**:

$A = 60 \times 15 - 2 \times 15^2$

$= 450$ Maximum area enclosed is 450 m^2.

---

1 A piece of rope is used to form a rectangle. Determine the largest area that can be formed with a 24-metre piece of rope.

2 A farmer wishes to form a temporary paddock using 100 metre of wire. If he uses an existing fence as one of the sides, determine the maximum area of the paddock.

3 Two numbers add up to 32. Find the numbers, if the sum of their squares is a minimum.

4 The total daily profit of a factory, which produces television sets, is given by $P = -\frac{3}{4}n^2 + 192n + 15$, where n is the number of television sets produced. Find the number of sets that must be produced to maximise profit and calculate the maximum profit.

5 Divide 150 into 2 non-zero integers x and y so that xy^2 is a maximum.

6 A manufacturer makes open cylindrical containers, using 400 cm^2 of aluminium sheeting. Determine the radius (to one decimal place), of the base of the cylinder, so that the volume will be a maximum. The setting out for this problem follows, but some steps have not been completed. Complete the missing steps.

$A = 2\pi rh + \pi r^2$ 1

$V = \pi r^2 h$ 2

$400 = 2\pi rh + \pi r^2$ substitute in 1

$2\pi rh =$ .... a)

$h =$ .... b)

$h = \frac{200}{\pi r} - \frac{r}{2}$

$V = \pi r^2 \times$ .... c) substitute in 2

$V =$ .... d)

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To determine where V is a maximum, we find the derivative and let $\frac{dV}{dr} = 0$

$\frac{dV}{dr} =$ .... e)

$\frac{d^2V}{dr^2} =$ .... f) (since $r > 0$ and $\frac{d^2V}{dr^2} < 0$, V is a maximum)

Let $\frac{dV}{dr} = 0$ i.e. $200 - \frac{3\pi r^2}{2} = 0$

$3\pi r^2 = 400$

$r =$ .... g)

$= 6.5$ (to 1 decimal place)

For a maximum volume open cylinder the radius = 6.5 cm.

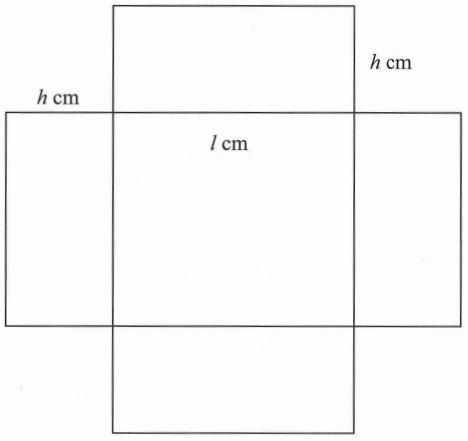
7 An aluminium can manufacturer wishes to make each can with a surface area of 600 cm^2.

If the cans have both a top and a bottom, determine (to 2 decimal places):

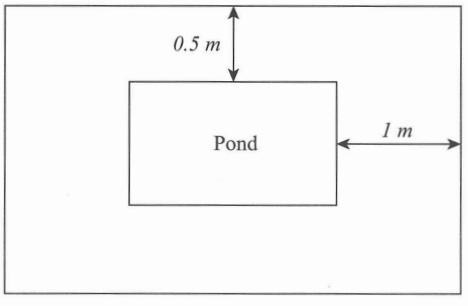
i) the radius of the can that will give the greatest volume

ii) the maximum volume of the can.
[Hint: Surface Area = $2\pi r^2 + 2\pi rh$.]

8 A manufacturer makes cartons (with no tops) that have square bases and rectangular sides. Assume the base length is l cm and the height is h cm, as in the diagram. If the volume of the box has to be 13 500 cm^3, determine the dimensions of the box that will give the least surface area.



9 A gardener creates an outdoor feature as shown below, which has an area of 98 cm^2. The feature comprises a pond surrounded by grass. The grass strip is 1 metre wide along two sides and 0.5 metre wide along the other two sides. Determine the dimensions of the feature so that the area of the pond is a maximum.



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Finding the primitive function is the inverse of differentiation. It is also referred to as "anti-differentiation".

The primitive function (anti-derivative) of ax^n is $\frac{ax^{n+1}}{n+1}+c$, where c is a constant, and it is often denoted by $F(x)$.

e.g.

1 Find the primitive function (anti-derivative) of:

i) $6x^4$ $\quad F(x)=\frac{6x^5}{5}+c$ or $\frac{6}{5}x^5+c$

ii) $3x^2+4x$ $\quad F(x)=x^3+2x^2+c$

iii) x^4+7x+4 $\quad F(x)=\frac{1}{5}x^5+\frac{7}{2}x^2+4x+c$

2 Given that $f'(x)=2x^2+3x$, and that $f(x)$ passes through the point (1,3), determine $f(x)$.

$f'(x)=2x^2+3x$

$f(x)=\frac{2}{3}x^3+\frac{3}{2}x^2+c$ and as the function passes through (1,3)

$3=\frac{2}{3}+\frac{3}{2}=c$

$c=\frac{5}{6}$

$f(x)=\frac{2}{3}x^3+\frac{3}{2}x^2+\frac{5}{6}$ Note: It is helpful to differentiate the answer as a check.

Find the primitive function (anti-derivative) of each of these functions.

1 $2x$ **2** $2x+8$ **3** $4x^3+x+1$ **4** $3x^2+8x-5$

5 $7-3x^2-3x^3$ **6** x^3-x^4 **7** x^4+x^6 **8** $x^{\frac{1}{2}}$

9 x^{-2} **10** x^{-3} **11** $2x^{-3}+2x+3$ **12** $x^{-\frac{1}{3}}+2x-4$

Find $f(x)$ for each of the following:

13 $f'(x)=6x+4x^3$ **14** $f'(x)=3x^2+2x^3$ **15** $f'(x)=4x^3+7x-\frac{1}{2}$

16 If $f'(x)=9x^2-4x+1$ and $f(x)$ passes through the point $(2,-5)$, find $f(x)$.

17 The gradient function of a curve is x^3+x and it passes through the point $(2,10)$. Find the equation of the curve.

18 Given that a curve passes through the point $(2,8)$, and $f'(x)=3x^2+7x-1$, then determine the equation of the function $f(x)$.

19 If $\frac{dy}{dx}=8x^3-3x^2-x$, and when $x=1$, $y=15$, determine an expression for y.

20 The gradient function of a curve is $4x^2-x^3+1$. If the curve also passes through the point $(2,0)$, find the equation of the curve.

21 For a given function, $\frac{d^2y}{dx^2}=2$. When $\frac{dy}{dx}=0$, $x=-2$ and when $x=3$, $y=21$. Find the equation of the function.

22 For a given function, $\frac{d^2y}{dx^2}=5$. At $\frac{dy}{dx}=2$, $x=1$ and at $x=2$, $y=1$. Find the equation of the function.

23 If $\frac{d^2y}{dx^2}=6x-6$ and $\frac{dy}{dx}=3$ when $x=2$, find y in terms of x, given that $x=1$ when $y=3$.

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$$\int ax^n\,dx = \frac{ax^{n+1}}{n+1} + c \quad \text{and} \quad \int (ax+b)^n\,dx = \frac{1}{a} \times \frac{(ax+b)^{n+1}}{n+1} + c$$

e.g. Find the following indefinite integrals:

1 $\int (3x^2 + 4x + 1)dx = x^3 + 2x^2 + x + c$

2 $\int \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right) dx = \int \left(x^{\frac{1}{2}} + x^{-\frac{1}{2}}\right) dx$

$$= \frac{2}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}}$$

$$= \frac{2}{3}\sqrt{x^3} + 2\sqrt{x} + c$$

3 $\int (6x-3)(x-5)\,dx = \int (6x^2 - 33x + 15)\,dx$

$$= 2x^3 - \frac{33}{2}x^2 + 15x + c$$

4 $\int \frac{x^3 - x}{x}\,dx = \int (x^2 - 1)\,dx$

$$= \frac{1}{3}x^3 - x + c$$

5 $\int (2x+3)^4 dx = \frac{1}{2} \times \frac{1}{5}(2x+3)^5 + c$

$$= \frac{1}{10}(2x+3)^5 + c$$

Determine each of the following indefinite integrals:

1 $\int (x+1)dx$

2 $\int (3x^2 - x)dx$

3 $\int dx$

4 $\int (4x^3 + 3x^2 - 2x)\,dx$

5 $\int (x^2 - 4x^3)dx$

6 $\int (x^3 - x^2 + 2x - 5)dx$

7 $\int (2x^3 - 4x^4 + x)\,dx$

8 $\int (x^{-2} - x^{-3} + 2)\,dx$

9 $\int (2x^{-3} + 2x^{-2})dx$

10 $\int (3x^2 + 4x^{-3} - x)dx$

11 $\int \left(x^{\frac{1}{2}} + x^{\frac{1}{3}}\right) dx$

12 $\int \left(2x^{\frac{1}{2}} - x^{\frac{2}{3}}\right) dx$

13 $\int \left(3x^{-2} + 4x^{-\frac{1}{3}}\right) dx$

14 $\int \left(\frac{2}{x^2} - \frac{2}{x^4} - 3\right) dx$

15 $\int (\sqrt{x} - \sqrt[3]{x} - 1)dx$

16 $\int \left(\frac{4}{\sqrt{x}} - 2\right) dx$

17 $\int \left(\frac{3}{\sqrt[3]{x}} - \frac{2}{\sqrt[5]{x}}\right) dx$

18 $\int (\sqrt[3]{x^2} + \sqrt{x^3})\,dx$

19 $\int \left(\frac{x^2 + x}{x}\right) dx$

20 $\int \left(\frac{3x^4 - x^3 + x^2}{x^2}\right) dx$

21 $\int \left(\frac{4x - 6x^3 + 3}{x^3}\right) dx$

22 $\int 3(x+3)^2 dx$

23 $\int 5(x-2)^4 dx$

24 $\int 6(2x-1)^2 dx$

25 $\int (x-5)^2 dx$

26 $\int (4-x)^3 dx$

27 $\int (4-3x)^3 dx$

28 $\int (4+2x)^4 dx$

29 $\int \sqrt{(x+7)}\,dx$

30 $\int \sqrt{(2x-1)}\,dx$

31 $\int \sqrt[3]{(x-1)}\,dx$

32 $\int \sqrt[3]{(9-3x)}\,dx$

33 $\int \frac{1}{\sqrt{2x-5}}\,dx$

34 $\int \frac{1}{\sqrt{4x-5}}\,dx$

35 $\int \frac{1}{\sqrt{6-5x}}\,dx$

36 $\int \frac{4}{\sqrt{2x-8}}\,dx$

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e.g. The following are examples of determinations of particular definite integrals:

1 $\int_{-1}^{2}(4x+3)dx=\left[2x^2+3x\right]_{-1}^{2}$

$=\{(2\times2^2+3\times2)-(2\times(-1)^2+3\times-1)\}$

$=15$

2 $\int_{1}^{4}\left(3t^{\frac{1}{2}}-3\right)dt=\left[2t^{\frac{3}{2}}-3t\right]_{1}^{4}$

$=\left\{2\times4^{\frac{3}{2}}-3\times4\right\}-\{2-3\}$ Note: $4^{\frac{3}{2}}=(\sqrt{4})^3$

$=5$

3 $\int_{1}^{2}\frac{x^4-x^2+1}{x^2}\,dx=\int_{1}^{2}(x^2-1+x^{-2})\,dx$

$=\left[\frac{x^3}{3}-x-\frac{1}{x}\right]_{1}^{2}$

$=\left\{\frac{2^3}{3}-2-\frac{1}{2}\right\}-\left\{\frac{1}{3}-1-1\right\}$

$=1\frac{5}{6}$

Find the following definite integrals:

1 $\int_{1}^{3}2x\,dx$

2 $\int_{-1}^{4}3x^2\,dx$

3 $\int_{-3}^{3}x^2\,dx$

4 $\int_{1}^{2}3\,dx$

5 $\int_{-1}^{1}dx$

6 $\int_{1}^{3}(x^2+1)\,dx$

7 $\int_{0}^{2}(x^2+4x-1)\,dx$

8 $\int_{-1}^{1}(x+3)(x-2)dx$

9 $\int_{1}^{3}(3-x)^2\,dx$

10 $\int_{-2}^{-1}(x^{-2}+x)\,dx$

11 $\int_{1}^{2}(x^{-2}-x^{-3})dx$

12 $\int_{1}^{2}\left(\frac{1}{x^2}+3\right)dx$

13 $\int_{-2}^{-1}\left(\frac{3}{x^3}+\frac{1}{x^2}\right)dx$

14 $\int_{-2}^{-1}x^2(x-3)\,dx$

15 $\int_{1}^{4}(\sqrt{x}-1)\,dx$

16 $\int_{1}^{2}\frac{dx}{x^3}$

17 $\int_{0}^{1}(x-1)^3dx$

18 $\int_{5}^{10}\sqrt{x-1}\,dx$

19 Determine the value of:

i) $\int_{-1}^{0}(x^2-7x+4)dx$ **ii)** $\int_{0}^{1}(x^2-7x+4)dx$ **iii)** $\int_{-1}^{1}(x^2-7x+4)\,dx$

Hence or otherwise find the value of **iv)** $\int_{-1}^{0}(x^2-7x+4)\,dx+\int_{0}^{1}(x^2-7x+4)\,dx.$

20 Find the value of $\int_{-2}^{-1}(x^2-3x+1)\,dx+\int_{-1}^{0}(x^2-3x+1)\,dx.$

21 Find the value of $\int_{-3}^{-1}(x^3-x^2)\,dx+\int_{-1}^{0}(x^3-x^2)\,dx.$

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Write down the primitive function (anti-derivative) of:

1 $6x^5+2x$ **2** $3x^2+4x+1$ **3** $5x^4+x$

4 x^3+4x-5 **5** x^4-2x+3 **6** 1

7 $x-3x^2$ **8** $2\sqrt{x}$ **9** $\frac{2}{x^3}+\frac{1}{x^2}$

10 $x^{-\frac{1}{3}}-x-4$ **11** $3(x-4)(x+1)$ **12** $\sqrt{x+1}$

13 A curve has a gradient function of $3x-4$. Determine the equation of the curve if it passes through the point $\left(1,2\frac{1}{2}\right)$.

14 Given $\frac{dy}{dx}=12x^3-4x$, and that when $x=-1$, $y=7$, find the equation of y in terms of x.

Determine the following indefinite integrals:

15 $\int(2x-1)\,dx$ **16** $\int(x^3+6x-5)\,dx$

17 $\int(5x^4-x^3-4)\,dx$ **18** $\int(\frac{1}{5}x^4-3x+5)\,dx$

19 $\int(x-4x^2+3)dx$ **20** $\int(3x-4)(3x+4)dx$

21 $\int x(4x^2-3x+1)\,dx$ **22** $\int(x-4)^2\,dx$

23 $\int(\sqrt{x}-3x)dx$ **24** $\int\left(\frac{1}{3\sqrt{x}}+4\right)dx$

Find the value of the following definite integrals:

25 $\int_0^2(4x^3+3x^2-4)\,dx$ **26** $\int_{-1}^0(6x^5-6x^2+2)\,dx$

27 $\int_0^3(x^2+4x-5)\,dx$ **28** $\int_{-1}^0(2x-4x^2-5x^4)dx$

29 $\int_{-1}^1(2x-3)(x-5)\,dx$ **30** $\int_1^2\left(\frac{1}{x^3}-x^{-2}\right)dx$

31 $\int_1^3 3(x-4)^2dx$ **32** $\int_0^4 3(3-x)^3dx$

33 For the curve $y=x^3-6x^2$, determine the stationary points and state their nature. Find the coordinates of the point of inflection. Draw a neat sketch of the curve.

34 Draw a neat sketch of the function $y=\frac{1}{3}x^3-x^2-3x+5$, showing important features.

35 The cost of manufacturing caravans is given by $P=2n^2-32n+151$, where P represents the production costs in \$10 000s and n represents the number of caravans manufactured. Determine the number of caravans that must be manufactured to minimise the production costs. Hence, calculate this minimum cost per caravan.

36 A tank manufacturer produces fully enclosed tanks, which are in the shape of rectangular prisms. The length of the tank is twice the width. Calculate the minimum surface area of the tank if the volume of the tank is 72 m^3.

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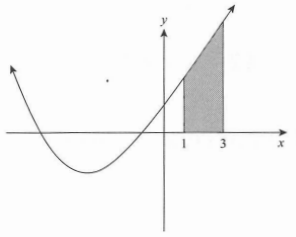
$Area = \int_a^b f(x)\,dx$ or $Area = \int_a^b y\,dx$ When determining the area between a curve and the x-axis, it is helpful to sketch the curve first, to determine whether any sections of the curve are below the axis.

e.g.

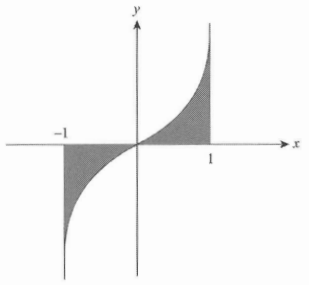
1 Find the area bounded by the curve $y = 3x^2 + 4x + 4$, the x-axis and the lines $x = 1$ and $x = 3$.



$$\begin{aligned} \text{Area} &= \int_1^3 (3x^2 + 4x + 4)\,dx \\ &= \left[x^3 + 2x^2 + 4x\right]_1^3 \\ &= \left\{(3^3 + 2\times 3^2 + 4\times 3) - (1^3 + 2\times 1^2 + 4\times 1)\right\} \\ &= 50 \text{ units}^2 \end{aligned}$$

If the curve is below the axis this will result in the integral being negative and the absolute value of the integral must be taken. If only part of the curve is negative the integral should be split into parts and added.

2 Find the area between the curve $y = x^3$, the x-axis and the lines $x = -1$ and $x = 1$.



$$\begin{aligned} \text{Area} &= \left|\int_{-1}^0 x^3 dx\right| + \int_0^1 x^3 dx \\ &= \left|\left[\frac{1}{4}x^4\right]_{-1}^0\right| + \left[\frac{1}{4}x^4\right]_0^1 \\ &= \left|\left\{0 - \left(\frac{1}{4}\times(-1)^4\right)\right\}\right| + \left\{\frac{1}{4}(1)^4 - 0\right\} \\ &= \left|-\frac{1}{4}\right| + \frac{1}{4} \\ &= \frac{1}{2} \text{ units}^2 \end{aligned}$$

Note: The integral $\int_{-1}^1 x^3 dx$ is zero – different to the area under the curve.

Determine the area bounded by:

1 the curve $y = x^2$ and the lines $x = 1$ and $x = 2$.

2 the curve $y = 3x^2 + 4x$ and the lines $x = 2$ and $x = 3$.

3 the curve $y = x^3$ and the lines $x = 0$ and $x = 2$.

4 Find the area bounded by the curve $y = x^2 - 2x$, the x-axis and the ordinates $x = 1$ and $x = 2$.

5 Show that the curve $y = -x^2 - 2x + 3$ cuts the x-axis at $x = 1$ and $x = -3$. Hence determine the area between the curve and the x-axis between the lines $x = 0$ and $x = 2$.

6 Find the area bounded by $y = x^2 + 3x + 2$, the x-axis and the ordinates $x = -3$ and $x = -2$.

7 Find the area between the curve $y = x^3$, the x-axis and the lines $x = -2$ and $x = 2$.

8 Calculate the area between the curve $y = x^2 + 4x - 5$, the x-axis and lines $x = -1$ and $x = 2$.

9 Determine the area between the x-axis and the curve $y = 4 - x^2$, from $x = 0$ and $x = 4$.

By first finding the x-intercepts, determine the area between the following curves and the x-axis:

10 $y = 1 - x^2$ **11** $y = x^2 - x$ **12** $y = x^2 + 5x - 6$

13 Determine the area of the region bounded by the curve $y = x^2 - 3x$ and the x-axis.

14 Determine the area of the region bounded by the curve $y = -x^2 + 5x - 4$ and the x-axis.

15 Find the area bounded by the curve $y = x(x-1)(x+2)$ and the x-axis.

16 Show that the function $y = x^3 - 4x$ can be written $y = x(x-2)(x+2)$.
Hence determine the area bounded by the curve $y = x^3 - 4x$ and the x-axis.

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To determine the area between a curve and the y-axis, first ensure the function is in the form $x = f(y)$ and then integrate with respect to y.

$$Area = \int_a^b f(y)\,dy \qquad \text{or} \qquad Area = \int_b^a x\,dy$$

e.g.

1 Determine the area in the first quadrant, between the curve $y = x^2$, the y-axis and lines $y = 1$ and $y = 4$.

$y = x^2$, thus $x = \sqrt{y} = y^{\frac{1}{2}}$

$$Area = \int_1^4 y^{\frac{1}{2}}\,dy$$
$$= \left[\frac{2}{3}y^{\frac{3}{2}}\right]_1^4$$
$$= \frac{2}{3}\left(4^{\frac{3}{2}} - 1^{\frac{3}{2}}\right)$$
$$= 4\frac{2}{3}\ units^2$$

Note: A sketch may be helpful.

2 By first finding the x-value of the point of intersection of the two curves, calculate the area between the curves $y = x + 2$ and $y = x^2$.

$y = x + 2$
$y = x^2$
Thus $x^2 = x + 2$
$x^2 - x - 2 = 0$
$(x - 2)(x + 1) = 0$
$x = 2,\ 1$

$$Area = \int_{-1}^{2} (x+2)\,dx - \int_{-1}^{2} x^2$$
$$= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3}\right]_{-1}^{2}$$
$$= 4\frac{1}{2}\ units^2$$

Note: Absolute value is taken when you are unsure of the relative positions of the curves.

---

1 Calculate the area in the first quadrant between $y = x^2$, the y-axis and the lines $y = 0$ and $y = 9$.

2 Calculate the area in the first quadrant between $x = y^{\frac{1}{3}}$, the y-axis and the lines $y = 1$ and $y = 8$.

Find the area between the following curves, the y-axis and the lines indicated.

3 $x = y^2 + 2$ between $y = 2$ and $y = 4$

4 $y = \sqrt{x - 3}$ between $y = 4$ and $y = 5$

5 $y^2 = 9x$ between $y = 2$ and $y = 3$

6 $y = \frac{1}{8}x^3$ between $y = 0$ and $y = 1$

7 $y = \frac{1}{x^2}$ between $y = 1$ and $y = 4$

8 To determine the area between the curve $y = x^3$ and $y = 4x$, the points of intersection are needed. Equating x^3 and $4x$ leads to $x(x^2 - 4) = 0$ with $x =$ ..... **i)**, ...**ii)**, or ..... **iii)**. Hence determine the area between the two curves in the first quadrant.

9 Show that $y = x^2$ and $y = 2 - x$ intersect at $x = -2$, and $x = 1$, and hence determine the area bounded by the two curves.

10 Show that $y = x^3$ and $y = 9x$ intersect at 3 points and hence determine the total area bounded by the two curves.

Determine the area between the curves. (Hint: Take care subtracting the second function.)

11 $y = x^4$ and $y = x$

12 $y = x^4$ and $y = -8x$

13 $y = -x^2 + 4$ and $y = x + 2$

14 $y = x^2$ and $y = -x^2 + 8$

15 $y = -x^2$ and $y = x^2 - 18$

16 $y = -x^2$ and $y = -3 - 2x$

17 Determine which area is greater, and by how much:

i) area between $y = x^2 + 2$, the x-axis and the lines $x = 1$ and $x = 2$

ii) area, in the first quadrant, between $y = x^2 + 2$, the y-axis and $y = 3$ and $y = 6$.

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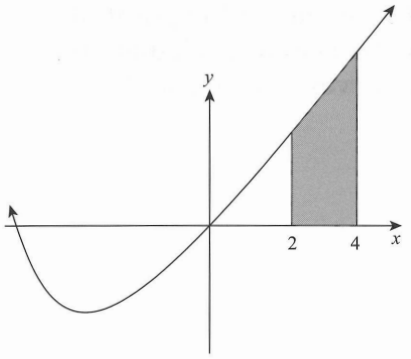
When it is difficult or impossible to integrate to find the area under a curve, an approximation method has to be used. Simpson's Rule and the Trapezoidal Rule are two approximation methods. In the example below, only one sub-interval has been used. Better approximations are obtained by using more sub-intervals.

e.g. Find the approximate area under the curve $y = x^2 + 4x - 1$, between the lines $x = 2$ and $x = 4$.



Solution using Simpson's Rule

$$A \approx \frac{b-a}{6}\left\{f(a)+4f\left(\frac{a+b}{2}\right)+f(b)\right\}$$

$$\approx \frac{4-2}{6}\left\{f(2)+4f\left(\frac{2+4}{2}\right)+f(4)\right\}$$

$$\approx 40\frac{2}{3}\ \text{units}^2$$

Solution using the Trapezoidal Rule

$$A \approx \frac{b-a}{2}\{f(a)+f(b)\}$$

$$\approx \frac{4-2}{2}\{f(2)+f(4)\}$$

$$\approx 42\ \text{units}^2$$

Note: **1** Simpson's Rule will give the exact value when the function is a parabola.

2 Both the above methods can be used to find the approximate value of $\int_2^4 (x^2 + 4x - 1)\,dx$.

3 There are different forms of these rules. Use the notation with which you are familiar.

Use one application of Simpson's Rule to find an approximation for these areas.

1 Find the area under the curve $y = x^2$, between the lines $x = 1$ and $x = 2$.

2 Find the area under the curve $y = x^2 + 1$, between the lines $x = 1$ and $x = 2$.

3 Find the area under the curve $y = x^3$, between $x = 1$ and $x = 3$.

Use Simpson's Rule and 2 sub-intervals to find an approximation for:

4 $\int_0^4 (x^3 + 1)\,dx$ **5** $\int_{-4}^0 (x^3 + 2x)\,dx$ **6** $\int_{-2}^0 (x^2 + 4x)\,dx$

7 Use one application of the Trapezoidal Rule to find an approximation for the area under the curve $y = \frac{1}{x}$, between $x = 1$ and $x = 3$.

8 Use one application of the Trapezoidal Rule to find an approximation for the area under the curve $y = x^2 + 4x + 1$, between $x = 0$ and $x = 2$.

Use the Trapezoidal Rule with 2 sub-intervals to find an approximation for each of these integrals:

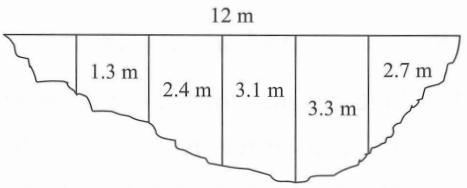
9 $\int_0^2 (x^3 + x^2)\,dx$ **10** $\int_1^3 \frac{1}{x+1}\,dx$

11 It is known that Simpson's Rule gives the exact value for the area under a curve when the curve is a parabola. Determine the percentage error, when the area under the curve $y = x^2 - 1$ between $x = 1$ and $x = 2$ is approximated using one application of the Trapezoidal Rule.

12 Using the values in the table, and the Trapezoidal Rule with 3 sub-intervals, give an approximation for $\int_1^4 f(x)\,dx$.

x	1	2	3	4
$f(x)$	9.3	8.4	8.1	7.9

13 The diagram at right shows the cross-section of a creek. Depths at equal intervals across the section are shown. Use Simpson's Rule and 7 function values to determine the area of this cross-section.



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$$\frac{d}{dx}(e^{f(x)}) = f'(x) \times e^{f(x)} \qquad \frac{d}{dx}\{\log_e f(x)\} = \frac{f'(x)}{f(x)}$$

Note: Remember to apply the properties of logarithms to simplify derivatives.

e.g.

1 Differentiate the following with respect to x:

i) e^{4x}

$\frac{d}{dx}(e^{4x}) = 4e^{4x}$

ii) $y = e^{2-3x}$:

$\frac{dy}{dx} = -3e^{2-3x}$

iii) $y = \log_e(3x+4)$

$\frac{dy}{dx} = \frac{3}{3x+4}$

2 Find the derivatives of the following functions:

i) $y = \log[(x^2+1)(x+1)]$

$= \log(x^2+1) + \log(x+1)$

$\frac{dy}{dx} = \frac{2x}{x^2+1} + \frac{1}{x+1}$

ii) $f(x) = \log\left(\frac{x-3}{2x+1}\right)$

$= \log(x-3) - \log(2x+1)$

$f'(x) = \frac{1}{x-3} - \frac{2}{2x+1}$

iii) $y = \log(6x+3)^3$

$= 3\log(6x+3)$:

$y' = 3 \times \frac{6}{6x+3}$

$= \frac{6}{2x+1}$

Differentiate with respect to x:

1 e^{6x}

2 e^{4x-1}

3 e^{-3x}

4 $f(x) = \frac{1}{4}e^x$

5 $f(x) = 3 + 2e^{x^3}$

6 $f(x) = x + \frac{1}{4}e^{4x}$

7 $y = 1 - e^{2x-3}$

8 $y = 2 - e^{x^2+1}$

9 $y = 2x - e^{4-5x}$

10 $y = e^{2x} + e^{3x}$

11 $y = e^{-x^2} - e^{-x}$

12 $y = x + e^{3x-4}$

13 $\log_e 5x^2$

14 $\log_e 8x$

15 $\log_e(2x-5)$

16 $y = \log_e(4x^2-1)$

17 $y = 3\log_e(2x-1)$

18 $y = \frac{1}{3}\log_e(6x-1)$

19 $y = \log_e\left(\frac{x+2}{x-1}\right)$

20 $y = \log_e\left(\frac{2x+3}{2x-1}\right)$

21 $y = \log_e\left(\frac{4x-3}{2x+1}\right)$

22 $\log_e[(x+3)(x-2)]$

23 $\log_e[(x-3)(x+4)]$

24 $\log_e[(2x-1)(2x+4)]$

25 $y = \log_e(x+3)^2$

26 $y = \log_e(2x-1)^3$

27 $y = \log_e(3-4x)^2$

28 $\log_e[(2x-1)^2(3x-2)]$

29 $y = \log_e\sqrt{x-1}$

30 $y = \log_e\sqrt{2x+1}$

31 $y = \log_e\sqrt[3]{1-3x}$

32 $f(x) = \log_e\frac{1}{x}$

33 $f(x) = \log_e\frac{1}{2x-5}$

34 $f(x) = \log_e\frac{2}{3x-4}$

35 $f(x) = \log_e\frac{1}{\sqrt{x-3}}$

36 $f(x) = \log_e\frac{1}{\sqrt{7-2x}}$

37 $f(x) = \log e\frac{1}{\sqrt[3]{3x+1}}$

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e.g.

1 Since $\frac{d}{dx}(e^{3x-2}) = 3e^{3x-2}$

then $\int 3e^{3x-2}\,dx = e^{3x-2} + c$

2 Since $\frac{d}{dx}\log_e(3x+4) = \frac{3}{3x+4}$

then $\int \frac{3}{3x+4}\,dx = \log_e(3x+4) + c$

3 $\int e^{4x-2}\,dx = \frac{1}{4}\int 4e^{4x-2}\,dx$

$= \frac{1}{4}e^{4x-2} + c$

Note: Make sure the coefficient of e is the derivative of the power.

4 $\int \frac{1}{6x-1}\,dx = \frac{1}{6}\int \frac{6}{6x-1}\,dx$

$= \frac{1}{6}\log_e(6x-1) + c$

Before integration make sure that the numerator is the derivative of the denominator.

5 $\int \frac{x+2}{x-2}\,dx = \int \frac{(x-2)+4}{x-2}\,dx$

$= \int \frac{x-2}{x-2} + \frac{4}{x-2}\,dx$

$= \int 1\,dx + 4\int\left(\frac{1}{x-2}\right)dx$

$= x + 4\log_e(x-2) + c$

Find each of these integrals.

1 $\int e^x\,dx$ **2** $\int 3e^{3x}\,dx$ **3** $\int -4e^{1-4x}\,dx$

4 $\int e^{2x}\,dx$ **5** $\int e^{4x+1}\,dx$ **6** $\int e^{3-2x}\,dx$

7 $\int 2xe^{x^2+1}\,dx$ **8** $\int 3x^2e^{x^3+1}\,dx$ **9** $\int xe^{x^2-5}\,dx$

10 $\int xe^{-3x^2}\,dx$ **11** $\int \frac{2}{2x+1}\,dx$ **12** $\int \frac{7}{7x+4}\,dx$

13 $\int \frac{1}{3x+4}\,dx$ **14** $\int \frac{1}{9x}\,dx$ **15** $\int \frac{1}{5x-1}\,dx$

16 $\int \frac{1}{2-9x}\,dx$ **17** $\int \frac{2x}{x^2-1}\,dx$ **18** $\int \frac{6x}{3x^2-4}\,dx$

19 $\int \frac{x}{x^2+2}\,dx$ **20** $\int \frac{3}{4x-1}\,dx$ **21** $\int \frac{x}{2-3x^2}\,dx$

22 $\int \frac{3x}{1-4x^2}\,dx$ **23** $\int \frac{2x^2+4x-1}{x}\,dx$ **24** $\int \frac{x^2-7x-3}{x}\,dx$

(Hint: Simplify first.)

25 Complete $\frac{x-2}{x+2} = \frac{x+2}{x+2} - \frac{\ldots\ldots}{x+2} = 1 - \frac{\ldots\ldots}{x+2}$. Hence find the integral $\int \frac{x-2}{x+2}\,dx$

26 Show that $\frac{x-1}{x+2} = 1 - \frac{3}{x+2}$, and hence find the integral $\int \frac{x-1}{x+2}\,dx$

27 Given $\frac{2x-13}{x-7} = \frac{2x-14}{x-7} + \frac{1}{x-7}$, find the integral $\int \frac{2x-13}{x-7}\,dx$

28 Show that $\frac{2x}{2x-1} = 1 + \frac{1}{2x-1}$, and hence find the integral $\int \frac{2x}{2x-1}\,dx$

29 Find $\int \frac{x-4}{x-1}\,dx$ **30** Find $\int \frac{2x+4}{2x+1}\,dx$

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The product rule, the quotient rule and function of a function rule (chain rule) are required in the following questions. Note: $\log_e(x) = \ln x$.

Differentiate the following:

1 $3e^x$

2 $2xe^{2x+1}$

3 $x^2e^{x^2}$

4 $2x\ln(x+1)$

5 $x^2\ln(x^2+1)$

6 $\dfrac{x}{e^x}$

Find the gradient function of the following:

7 $y = e^{x+1}\ln x$

8 $y = e^{2x}\ln(x+3)$

9 $y = \dfrac{2x}{e^{2x}}$

10 $y = \dfrac{\ln x}{e^x}$

Determine the gradient of the tangent to the curve:

11 $y = \ln x$ at the point where $x = 1$

12 $y = \ln\left(\dfrac{1}{2x-1}\right)$ at the point where $x = 2$

13 $y = e^{4x}$ at the point where $x = 1$

14 e^{1-x^2} at the point where $x = 1$.

Find the following definite integrals to 2 decimal places:

15 $\int_1^3 \dfrac{1}{1+x}dx$

16 $\int_2^3 \dfrac{1}{1+2x}dx$

17 $\int_1^2 \dfrac{x}{1+x^2}dx$

18 $\int_0^1 \dfrac{x^2}{1+x^3}dx$

19 $\int_0^1 e^{x+1}dx$

20 $\int_1^2 4xe^{x^2-1}dx$

21 $\int_0^1 \dfrac{2e^{2x}}{e^{2x}+1}dx$

22 $\int_1^2 \dfrac{5e^{3x}}{e^{3x}+1}dx$

23 Find the area of the region bounded by the curve $y = 2e^x$, the x-axis and the lines $x = 1$ and $x = 3$. (Leave the answer in terms of e.)

24 Find the area, in terms of e, bounded by $y = e^{-3x}$, the x-axis and the lines $x = -2$ and $x = 0$.

25 Determine the area of the region bounded by the curves $y = e^{2x}$, $y = e^{3x}$, and the line $x = 1$.

26 Determine the exact area bounded by $y = \dfrac{1}{x}$, the x-axis and the ordinates $x = 2$ and $x = 4$.

27 Calculate the area of the region bounded by $y = \dfrac{1}{x+1}$, the x-axis, and the ordinates $x = 2$ and $x = 4$.

28 Calculate the area of the region bounded by the curve $y = \dfrac{1}{x} + x$, the x-axis and the ordinates $x = 2$ and $x = 4$.

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$\frac{d}{dx}(\sin x) = \cos x$ $\frac{d}{dx}(\cos x) = -\sin x$ $\frac{d}{dx}(\tan x) = \sec^2 x$

Note: Angles are measured in radians.

e.g.

1 $\frac{d}{dx}(4\cos x) = -4\sin x$

2 $\frac{d}{dx}\cos(x^2+4) = -2x\sin(x^2+4)$ (chain rule)

3 $\frac{d}{dx}\left(\frac{x}{\sin x}\right) = \frac{\sin x - x\cos x}{\sin^2 x}$ (quotient rule)

4 $\frac{d}{dx}(x^2\tan x) = x^2\sec^2 x + 2x\tan x$ (product rule)

Differentiate the following with respect to x:

1 $3\sin x$

2 $4\cos x$

3 $9\tan x$

4 $3\cos x - 2\tan x$

5 $\frac{1}{2}(5\cos x + 2\sin x)$

6 $\frac{1}{2}\sin x - \frac{1}{4}\cos x - 6\tan x$

7 $\cos 4x$

8 $\tan 6x$

9 $\sin\frac{2}{3}x$

10 $4\tan 3x$

11 $\frac{1}{3}\cos 3x$

12 $\frac{1}{2}\sin^2 x$

13 $\cos^2 2x$

14 $\tan^2 3x$

15 $\log(\cos x)$

16 $\log(\sin x)$

17 $x^2\sin x$

18 $x^3\cos 3x$

19 $2x^2\tan 4x$

20 $e^{2x}\sin x$

21 $\frac{1}{\cos x}$

22 $\frac{1}{2\sin 2x}$

23 $\frac{x}{\cos x}$

24 $\frac{x^2}{\sin x}$

25 Find the gradient function of $y = 2\tan x$.

26 Find the gradient function of $y = x\cos 2x$.

27 Find the exact gradient of the curve $y = \sin x$ at the point where $x = \frac{\pi}{4}$.

28 Find the gradient of the curve $y = \cos x$ at the point where $x = \frac{\pi}{6}$.

29 Find the gradient of the tangent to the curve $y = \tan x$ at the point where $x = \frac{\pi}{4}$.

30 If $y = 3\sin\theta + 4\cos\theta$, find $\frac{dy}{dx}$ when $\theta = \frac{\pi}{3}$.

31 Find the maximum and minimum values of the function $y = \sin x + \cos x$.

32 Find the maximum and minimum values of the function $y = \cos x + \sqrt{3}\sin x$.

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$\frac{d}{dx}\sin(x) = \cos x$ $\quad$ $\frac{d}{dx}\cos(x) = -\sin x,$ $\quad$ $\frac{d}{dx}\tan(x) = \sec^2 x$

$\int \cos x\,dx = \sin x + c$ $\quad$ $\int \sin x\,dx = -\cos x + c$ $\quad$ $\int \sec^2 x\,dx = \tan x + c$

Also: $\int \sin ax\,dx = -\frac{1}{a}\cos ax + c$ $\quad$ $\int \cos ax\,dx = \frac{1}{a}\sin ax + c$ $\quad$ $\int \sec^2 ax\,dx = \frac{1}{a}\tan ax + c$

e.g.

1 $\int \sin 3x\,dx = -\frac{1}{3}\cos 3x + c$ $\quad$ **2** $\int \cos(3x+4)\,dx = \frac{1}{3}\sin(3x+4) + c$

Integrate these functions with respect to x:

1 $4\sin x$ $\quad$ **2** $2\cos x$ $\quad$ **3** $3\sec^2 x$

4 $\sin 3x$ $\quad$ **5** $\cos 4x$ $\quad$ **6** $-3\sin 5x$

7 $5\sin 4x$ $\quad$ **8** $3\cos 2x$ $\quad$ **9** $2\sec^2 4x$

10 $6\cos 4x$ $\quad$ **11** $\sin\frac{x}{4}$ $\quad$ **12** $-\cos\frac{x}{3}$

13 $\frac{1}{3}\sin\frac{x}{3}$ $\quad$ **14** $-\frac{2}{3}\cos 5x$ $\quad$ **15** $\cos(2x-4)$

16 $\sin(1-3x)$ $\quad$ **17** $\sin 2x + \frac{1}{2}\cos 2x$ $\quad$ **18** $4\cos 2x - 3\sec^2 4x$

19 Given that $\tan^2 x + 1 = \sec^2 x$, find $\int \tan^2 x\,dx$.

Find the exact value of the following definite integrals:

20 $\int_0^{\frac{\pi}{2}} \cos x\,dx$ $\quad$ **21** $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \sin x\,dx$ $\quad$ **22** $\int_0^{\frac{\pi}{6}} \sec^2 x\,dx$

23 $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos\theta\,d\theta$ $\quad$ **24** $\int_0^{\frac{\pi}{4}} (2\sin x - 5\sec^2 x)\,dx$ $\quad$ **25** $\int_0^{\frac{\pi}{2}} (-3\cos x + 1)\,dx$

26 $\int_0^{\frac{\pi}{8}} \sec^2 2\theta\,d\theta$ $\quad$ **27** $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sec^2\frac{x}{2}\,dx$ $\quad$ **28** $\int_0^{\frac{\pi}{2}} (2-\sin 2x)\,dx$

29 The gradient function of a curve is $3\sin x$. If it passes through the point $\left(\frac{\pi}{3}, \frac{5}{2}\right)$, what is the equation of the curve?

30 The gradient function of a curve is $\frac{1}{2}\cos x$. Determine the equation of the curve if it passes through the point $\left(\frac{\pi}{4}, 0\right)$.

31 Find the area enclosed by the curve $y = \cos x$ and the x-axis from $x = \frac{\pi}{2}$ to $x = \frac{3\pi}{2}$.

32 Find the area enclosed by the curve $y = \sin x$ and the x-axis from $x = 0$ and $x = \frac{\pi}{4}$.

33 Find the area between $y = 2\sin x$ and the x-axis from $x = -\frac{\pi}{3}$ to $x = \frac{\pi}{6}$.

34 Find the area between $y = 1 + \cos x$ and the x-axis from $x = 0$ to $x = \frac{\pi}{4}$.

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A solid of revolution can be formed by the rotation of a curve about the x-axis or the y-axis.

About the x-axis, the volume is found by:

$$V = \pi \int_a^b f(x)^2 dx \text{ or } V = \pi \int_a^b y^2 dx$$

About the y-axis, the volume is found by:

$$V = \pi \int_a^b f(y)^2 dy \text{ or } V = \pi \int_a^b x^2 dy$$

e.g.

1 Determine the volume of the solid formed when $y = \sqrt{x}$ is rotated about the x-axis from $x = 1$ to $x = 3$.

$$V = \pi \int_1^3 (\sqrt{x})^2 dx$$
$$= \pi \int_1^3 x\, dx$$
$$= \pi \left[\frac{x^2}{2} \right]_1^3$$
$$= \pi \left\{ \frac{9}{2} - \frac{1}{2} \right\}$$
$$= 4\pi \text{ units}^2$$

2 Determine the volume of the solid formed when $y = x^2$ is rotated about the y-axis from $y = 0$ to $y = 3$.

$$V = \pi \int_0^3 y\, dy$$
$$= \left\{ \frac{9}{2} - 0 \right\}$$
$$= \frac{9\pi}{2} \text{ units}^2$$

Find the volume of the solid of revolution when the curves are rotated about the x-axis between the x-values indicated. (Leave answers in terms of π.)

1 $y = x^2$ between $x = 0$ and $x = 2$.

2 $y = x^3$ between $x = 1$ and $x = 2$.

3 $y = x^2 + 1$ between $x = 0$ and $x = 2$.

4 Name the solid formed when $y = x$ is rotated about the x-axis from $x = 0$ to $x = 3$. Determine its volume.

5 Find the volume of the solid of revolution when the area between $y = x(x-1)$ and the x-axis is rotated about the x-axis. (Answer to 2 decimal places.)

6 Find the volume of the sphere formed when the semi-circle $y = \sqrt{4 - x^2}$ is rotated about the x-axis.

7 Find the sphere volume formed when the semi-circle $y = \sqrt{1 - x^2}$ is rotated about the x-axis.

8 Find the volume of the solid generated when the area between the y-axis, $y = 1$, $y = 3$ and the curve $y = x^2 + 1$ is rotated about the y-axis.

9 Find the volume of the solid generated when $y = x^2 - 1$ is rotated about the y-axis between $y = -1$ and $y = 3$.

10 Find the volume of the solid generated when $y = x^2 + 2$ is rotated about the y-axis from $y = 2$ to $y = 4$.

11 Find the volume of the solid formed when the area between the curve $xy = 1$ and the lines $y = 1$ and $y = 4$ is rotated about the y-axis.

12 Find the volume of the solid of revolution when the area between $y = e^x$ and the lines $x = 1$ and $x = 3$ is rotated about the x-axis.

13 Find the volume of the solid of revolution when the area enclosed by $y = \sec x$, $x = \frac{\pi}{6}$ and $x = \frac{\pi}{3}$ is rotated about the x-axis.

14 Show that the curves $y = x^2$ and $y = x + 2$ intersect at the points (–1, 1) and (2, 4). The area between these two curves is rotated about the x-axis. The volume of the solid generated is given by $V = \pi \int_{-1}^{2} (x+2)^2 dx - \pi \int_{-1}^{2} (x^2)^2 dx$. Determine the volume in terms of π.

15 Find the volume when the region bounded by $y = x^2$ and $y = 2x$ is rotated about the x-axis.

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Velocity is defined as the rate of change of displacement. If $x = f(t)$ then $v = \frac{dx}{dt}$ or $v = \dot{x}$.

Acceleration is the rate of change of velocity. If $v = f(t)$ then $a = \frac{dv}{dt}$ or $a = \frac{d^2x}{dt^2}$ or $a = \dot{v} = \ddot{x}$.

e.g. A particle moves in a straight line so that after t seconds, its displacement, x metres from a fixed point O, is given by $x = t^3 - 6t^2 + 9t + 6$. Determine:

i) its initial position

ii) its position, velocity and acceleration after 4 seconds

iii) when it first comes to rest.

i) Initial position occurs when t = 0 i.e. $x = 6$
The particle is initially 6 metres to the right of O.

ii) $v = 3t^2 - 12t + 9$ and $a = 6t - 12$

When t = 4:

$x = 4^3 - 6\times4^2 + 9\times4 + 6$ $\quad$ $v = 3\times4^2 - 12\times4 + 9$ $\quad$ $a = 24 - 12$

$= 10$ $\quad$ $= 9$ $\quad$ $= 12$

After 4 seconds, the particle is 10 metres to the right of O, with a velocity of 9 m/s and acceleration of 12 m/s^2.

iii) At rest when $v = 0$ $\quad$ $3t^2 - 12t + 9 = 0$

$t^2 - 4t + 3 = 0$

$(t-1)(t-3) = 0$

$t = 1,\ 3$ $\quad$ The particle first comes to rest after 1 second.

---

1 A particle is moving in a straight line so that after t seconds its position from a fixed point O is given by $x = t^3 - 3t^2 + 4t + 3$, where x is in metres. Determine:
i) its initial position **ii)** its position, velocity and acceleration after 4 seconds.

2 The position of a particle from a fixed point O as it moves along a straight line is given by $x = t^3 - 3t^2 - 9t + 4$. Show that the particle comes to rest only once and determine the position and acceleration when this occurs.

3 A particle is moving in a straight line so that its displacement from a fixed point O at t seconds is given by $x = 27t - t^3$.
i) Show that the particle comes to rest once and find the time when this occurs.
ii) How far has it travelled when it comes to rest?
iii) Show that the particle passes through O when $t = \sqrt{27}$ s and hence determine the velocity at this time.

4 A particle moves in a straight line and its position, in metres, from a fixed point O, is given by $x = t^2 + 5$. Determine:
i) its initial position **ii)** the time when its velocity is 15 m/s^2
iii) the distance travelled in the third second.

5 The position of a particle that is projected vertically upwards is given by $y = 7 + 6t - t^2$, where y is the height in metres above the ground and t is the time in seconds. Determine:
i) the particle's initial height **ii)** the maximum height reached
iii) the time taken to reach the ground and its velocity at that time.

6 A particle is moving in a straight line. The acceleration of the particle is 4 m/s^2. After 3 seconds, the velocity is 5 m/s and the particle is 2 metres to the left of the origin. Determine the position of the particle after 2 seconds.

7 The velocity of a particle that travels in a straight line is given by $v = 1 - \sqrt{2}\sin t,\ 0 \le t \le 2\pi$ where v is in m/s and t is in seconds.
i) Find the times when the particle is at rest
ii) What is the maximum velocity of the particle?
iii) If the particle is initially at the origin, determine an expression for its displacement.

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Differentiate the following:

1 e^{6x}

2 $4e^{3x^2+5}$

3 $\log_e(3x-5)$

4 $\log_e(7x^2-2)$

5 $\log_e((x+3)(x^2+4x))$

6 $\log_e\left(\frac{6x-1}{2x+4}\right)$

7 $\log_e(\frac{3x+4}{2x-1})$

8 $3x\ln(2x-1)$

9 $\frac{5x}{e^{3x}}$

10 $\sin 4x$

11 $\cos(x^2-3)$

12 $\frac{2}{3}\cos 3x$

13 $\frac{1}{2}\tan 4x$

14 $\sin^3 x$

15 $x\cos x$

16 $e^x \tan 3x$

Integrate the following:

17 $\int 4e^{2x}\,dx$

18 $\int \frac{3}{2x-1}dx$

19 $\int \frac{2x}{1+x^2}dx$

20 $\int 2xe^{2x^2+1}\,dx$

21 $\int 5\cos x$

22 $\int 4\sin x\,dx$

23 $\int 3\sin 2x\,dx$

24 $\int \cos 2x\,dx$

25 $\int \sec^2 3x\,dx$

26 $\int 3\sec^2 4x\,dx$

Find the value of the following definite integrals:

27 $\int_1^2 \frac{x^2}{1+x^3}dx$

28 $\int_0^{\frac{\pi}{6}} \sin 2x\,dx$

29 $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cot x\,dx$ $\left[\text{Hint : Let } \cot x = \frac{\cos x}{\sin x}.\right]$

30 Find $\int \frac{x+4}{x-3}dx$.

$\left[\text{Hint : First show that } \frac{x+4}{x-3} = \frac{x-3}{x-3} + \frac{7}{x-3}.\right]$

31 Find the indefinite integral $\int \frac{2x-1}{2x+3}\,dx$.

32 Determine the area of the region bounded by the curve $y = x^2 + x - 12$, the x-axis and the lines $x = 0$ and $x = 6$.

33 Calculate the area of the region in the first quadrant bounded by the curve $y = x^2$ and the line $y = 4 - 3x$.

34 Calculate the area of the region in the first quadrant, bounded by $y = x^2 + 4$, the y-axis and the lines $y = 5$ and $y = 8$.

35 Using Simpson's Rule with 3 function values, find the approximate value of $\int_1^3 \frac{1}{x+1}dx$.

36 Determine the volume of the solid of revolution when the area between $y = x^2 - 9$ and the x-axis is rotated about the x-axis.

37 Calculate the volume formed when the area bounded by $y = \sec x$ from $x = \frac{\pi}{4}$ to $x = \frac{\pi}{3}$ is rotated about the x-axis. Leave your answer in exact form.

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answers

1–limits

(1) 20 **(2)** 0 **(3)** 13 **(4)** 10 **(5)** –2 **(6)** 0 **(7)** –3
(8) –23 **(9)** 20 **(10)** 8 **(11)** –8 **(12)** –2 **(13)** 6 **(14)** 10
(15) –18 **(16)** 5 **(17)** 3 **(18)** –4 **(19)** 8 **(20)** –2
(21) $\frac{1}{4}$ **(22)** $\frac{3}{2}$ **(23)** 2 **(24)** 5 **(25)** $\frac{5}{6}$ **(26)** 4 **(27)** 1
(28) 6 **(29)** 0 **(30)** 0 **(31)** 0 **(32)** 2

2–differentiation from first principles

(1) 2 **(2)** 2 **(3)** 3 **(4)** –5 **(5)** $\frac{1}{3}$ **(6)** $4x$
(7) $6x$ **(8)** $4x$ **(9)** $-2x$ **(10)** $2x+1$ **(11)** $8x+4$
(12) $4-2x$ **(13)** $3x^2+3xh+h^2$, $3x^2$ **(14)** $3x^2$
(15) $3x^2+4$

3–basic differentiation

(1) 0 **(2)** 0 **(3)** 7 **(4)** 6 **(5)** $2x$ **(6)** $18x$ **(7)** $-14x$
(8) $4x^3$ **(9)** $-24x^3$ **(10)** $18x^2$ **(11)** $90x^9$ **(12)** $15x^2$
(13) $14x$ **(14)** $32x^3+6x$ **(15)** $12x-7$ **(16)** $27x^2+8$
(17) $12x^3+12x-1$ **(18)** $9x^2+4$ **(19)** –4 **(20)** $-28x$
(21) $3-14x$ **(22)** 7 **(23)** $6-6x$ **(24)** $-6x+12x^2$
(25) $-3x^{-4}$ **(26)** $-12x^{-3}$ **(27)** $-21x^{-4}+2x$ **(28)** $\frac{-1}{x^2}$
(29) $\frac{-8}{x^3}+1$ **(30)** $\frac{-6}{x^3}-\frac{3}{x^4}$ **(31)** $8x+20$ **(32)** $6x+12$
(33) $2x+7$ **(34)** $4x+1$ **(35)** $6x+5$ **(36)** $27x^2+2x+9$
(37) $\frac{-1}{x^2}$ **(38)** $\frac{-1}{x^2}-\frac{6}{x^3}$ **(39)** $\frac{-1}{x^2}+\frac{10}{x^3}$ **(40)** $\frac{-3}{x^4}-7$
(41) $1-\frac{1}{x^2}$ **(42)** $4x+\frac{1}{x^2}$ **(43)** 39, 63 **(44)** 13, 148
(45) –86, 40

4–gradient function and angle of inclination

(1) 7 **(2)** 6 **(3)** 4 **(4)** $2x+7$ **(5)** $3x^2-3$ **(6)** $5-18x$
(7) $6x+3$ **(8)** $-36x^2$ **(9)** $2x+7$ **(10)** $-x^{-2}$ **(11)** $\frac{-3}{x^2}$
(12) $\frac{-12}{x^3}$ **(13)** 12 **(14)** 6 **(15)** 16 **(16)** 5 **(17)** 1
(18) 21 **(19)** 7 **(20)** 10 **(21)** 87° 57′ **(22)** 85° 36′
(23) 45° **(24)** 86° 49′ **(25)** $\sqrt{3}$ **(26)** –1 **(27)** (3, –5)

5–equations of tangents and normals

(1) $6x-y-1=0$, $x+6y-31=0$ **(2)** $2x+y+1=0$, $x-2y-12=0$ **(3)** $x-y+2=0$, $x+y=0$
(4) $x+3y-6=0$, $3x-y-8=0$ **(5)** $x+y-4=0$, $x-y=0$
(6) $4x-y-2=0$, $x+4y-9=0$ **(7)** $15x-y-16=0$, $x+15y-212=0$ **(8)** $9x+y+8=0$, $x-9y+174=0$
(9) (3, 10), $6x-y-8=0$ **(10)** (2, 8), $12x-y-16=0$
(11) (3, –1), $x+2y-1=0$ **(12)** (1, –4), $x+y+3=0$

6–calculus review 1

(1) 6 **(2)** 7 **(3)** –3 **(4)** $10x$ **(5)** $6x+1$ **(6)** $14x-2$
(7) 13 **(8)** 11 **(9)** 6 **(10)** –4 **(11)** 10 **(12)** 4 **(13)** 16
(14) 7 **(15)** –7 **(16)** 3 **(17)** $\frac{2}{3}$ **(18)** 1 **(19)** 0 **(20)** $12x^2$ **(21)** $-35x^4$ **(22)** $5-14x-4x^3$ **(23)** $-12x^2-6x^5$
(24) $-4x^{-5}$ **(25)** $-4x+1-3x^{-4}$ **(26)** $12x^2-6+x^{-2}$
(27) $5x^4-2x^3+2x^{-3}$ **(28)** 7 **(29)** $-8+5x^4$ **(30)** $2+3x^{-4}$
(31) $-20x^{-6}$ **(32)** $\frac{-6}{x^3}$ **(33)** $\frac{-1}{x^2}$ **(34)** $\frac{-9}{x^4}-\frac{2}{x^3}$ **(35)** $\frac{16}{x^5}$
(36) $2x-2$ **(37)** $24x-54x^2$ **(38)** $8x-3-\frac{1}{x^2}$
(39) $2x-1-\frac{1}{x^2}$ **(40)** $7x-y+4=0$, $x+7y-28=0$
(41) $10x+y-16=0$, $x-10y-42=0$ **(42)** $x+y+4=0$, $x-y=0$ **(43)** $14x-y-10=0$, $x+14y-57=0$ **(44)** 72°
(45) 84° **(46)** 45° **(47)** (2, –1) **(48)** (–2, –3)

7–differentiation with rational powers

(1) $\frac{1}{2}x^{\frac{-1}{2}}$ **(2)** $\frac{1}{3}x^{\frac{-2}{3}}$ **(3)** $\frac{1}{5}x^{\frac{-4}{5}}$ **(4)** $\frac{2}{3}x^{\frac{-1}{3}}$ **(5)** $\frac{3}{2}x^{\frac{1}{2}}$
(6) $\frac{3}{4}x^{\frac{-1}{4}}$ **(7)** $x^{\frac{-1}{2}}$ **(8)** $\frac{9}{4}x^{\frac{-1}{4}}$ **(9)** $\frac{-1}{3}x^{\frac{-4}{3}}$ **(10)** $\frac{-2}{3}x^{\frac{-5}{3}}$
(11) $-3x^{\frac{-7}{4}}$ **(12)** $\frac{-16}{3}x^{\frac{-7}{3}}$ **(13)** $8x^7$ **(14)** $7x^6$ **(15)** $-x^{-2}$
(16) $\frac{-2}{x^3}$ **(17)** $\frac{3}{2}x^{\frac{-1}{2}}$ **(18)** $\frac{4}{3(\sqrt[3]{x^2})}$ **(19)** $\frac{1}{5(\sqrt[5]{x^3})}$
(20) $\frac{3\sqrt{x}}{2}$ **(21)** $\frac{5\sqrt{x^3}}{2}$ **(22)** $\frac{4\sqrt[3]{x}}{3}$ **(23)** $\frac{7\sqrt{x^5}}{2}$
(24) $\frac{7\sqrt[3]{x^4}}{3}$ **(25)** $\frac{1}{2\sqrt{x}}$ **(26)** $\frac{5\sqrt[4]{x}}{4}$ **(27)** $\frac{3\sqrt{x}}{2}$ **(28)** $\frac{1}{6\sqrt{x}}$
(29) $\frac{5\sqrt[3]{x^2}}{3}$ **(30)** $\frac{11\sqrt[3]{x^8}}{3}$ **(31)** $\frac{1}{8\sqrt{x}}$ **(32)** $\frac{-1}{5\sqrt[5]{x^6}}$
(33) $\frac{-1}{2\sqrt{x^3}}$ **(34)** $\frac{1}{15\sqrt[3]{x^2}}$ **(35)** $\frac{-2}{9\sqrt[3]{x^5}}$ **(36)** $\frac{-1}{9\sqrt[3]{x^4}}$
(37) $1+\frac{1}{2\sqrt{x}}$ **(38)** $\frac{1}{2\sqrt{x}}+8x$ **(39)** $2x-\frac{1}{3\sqrt[3]{x^4}}$
(40) $\frac{1}{2\sqrt{x}}+\frac{3}{2\sqrt{x^3}}$ **(41)** $\frac{1}{5\sqrt[5]{x^4}}+\frac{1}{x^2}$ **(42)** $\frac{1}{3\sqrt[3]{x^2}}+\frac{1}{3\sqrt[3]{x^4}}$
(43) $\frac{1}{4}$ **(44)** $\frac{1}{3}$

8–differentiation – product rule

(1) $6x^2+8x+4$ **(2)** $3x^2-10x+3$ **(3)** $8x^3-9x^2+8$
(4) $3x^2+6x-18$ **(5)** $10x^4+36x^3+4x+9$
(6) $15x^4-36x^3+27x^2-54x$ **(7)** $4x^3-6x^2-8x+8$
(8) $14x^6+20x^4-8x^3-8x$ **(9)** $7x^6-18x^5+24x^3-54x^2$
(10) $10x^4-4x^3-24x^2+8x$ **(11)** $8x^3+15x^2+8$
(12) $42x^2-30x-9$ **(13)** $5x^4+12x^2+12x$
(14) $5x^5+4x^3-3x^2-2x$ **(15)** $3x^2+18x+19$
(16) $10x^4+12x^3-18x^2-18x$ **(17)** $96x^3+108x^2-36x-27$
(18) $30x^4+6x^2-8$ **(19)** $-30x^4+32x^3-72x^2+64x$
(20) $150x^4+160x^3-18x^2-16x$ **(21)** $8x^3-12x^2+6x-5$
(22) $48x^3-51x^2+12x+4$ **(23)** 31 **(24)** 47 **(25)** 2
(26) –6 **(27)** 4 **(28)** –3 **(29)** $\frac{-1}{27}$ **(30)** $26x-y-48=0$

9–differentiation – quotient rule

(1) $\frac{1}{(2x+1)^2}$ **(2)** $\frac{-8}{(3x+2)^2}$ **(3)** $\frac{4x^2-10x}{(4x-5)^2}$ **(4)** $\frac{-6x^3-12x^2}{(3x+4)^2}$
(5) $\frac{1-x^2}{(x^2+1)^2}$ **(6)** $\frac{-3x^2-6}{(x^2-2)^2}$ **(7)** $\frac{6x^2}{(3x^2-x)^2}$ **(8)** $\frac{1}{(2x+1)^2}$
(9) $\frac{-7}{(4x-5)^2}$ **(10)** $\frac{x^2-4x-1}{(x-2)^2}$ **(11)** $\frac{x^2+2x+3}{(x+1)^2}$
(12) $\frac{2x^2-8x+6}{(4-2x)^2}$ **(13)** $\frac{-x^2-3}{x^4}$ **(14)** $\frac{x^4+3x^2}{(x^2+1)^2}$
(15) $\frac{24x^2+40x}{(6x+5)^2}$ **(16)** $\frac{-4x}{(x^2-1)^2}$ **(17)** $\frac{2x^4+3x^2}{(2x^2+1)^2}$
(18) $\frac{2x^3-1}{(x^3+1)^2}$ **(19)** $\frac{-1}{2}$ **(20)** –1 **(21)** 7° **(22)** 42°

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(23) $x-y+6=0$ **(24)** $11x+y-13=0$
(25) $8x+9y+7=0$ **(26)** $3x-4y-1=0,\ 8x+6y-11=0$

10–differentiation – chain rule

(1) $12(4x+2)^2$ **(2)** $20(5x+3)^3$ **(3)** $-24(7-6x)^3$

(4) $35(7x-3)^4$ **(5)** $6x(x^2+4)^2$ **(6)** $24x(4x^2-7)^2$

(7) $-36x^2(2-3x^3)^3$ **(8)** $60x^3(3x^4+6)^4$ **(9)** $-14x(7x^2-3)^{-2}$

(10) $-3(3x-1)^{-2}$ **(11)** $\dfrac{8x}{(3-4x^2)^2}$ **(12)** $\dfrac{-4x}{(2x^2+1)^2}$

(13) $\dfrac{2}{\sqrt{4x+2}}$ **(14)** $\dfrac{3}{2\sqrt{3x-5}}$ **(15)** $\dfrac{-2x}{\sqrt{3-2x^2}}$

(16) $-14(7x-3)^{-3}$ **(17)** $-\dfrac{1}{3}(6-x)^{-\frac{2}{3}}$ **(18)** $\dfrac{4x}{3\sqrt[3]{(2x^2-1)^2}}$

(19) $\dfrac{12}{\sqrt{6x-7}}$ **(20)** $\dfrac{-3}{\sqrt{6-2x}}$ **(21)** $\dfrac{6}{\sqrt{(2-4x)^3}}$ **(22)** -12

(23) 876 204 **(24)** $\dfrac{4}{\sqrt{15}}$ **(25)** 9 **(26)** $64, -\dfrac{1}{64}$

(27) $6x+y-7=0, x-6y+5=0$

11–calculus review 2

(1) $\dfrac{2}{3x^{\frac{1}{3}}}$ **(2)** $3x^2+6x-2$ **(3)** $\dfrac{2}{(x+1)^2}$ **(4)** $6(3+2x)^2$

(5) $\dfrac{-1}{(x+3)^2}$ **(6)** $\dfrac{7}{2\sqrt{x}}$ **(7)** $4x^3+2x$ **(8)** $\dfrac{-15}{(x-5)^2}$

(9) $-16(3-4x)^3$ **(10)** $\dfrac{-2x}{(x^2-3)^2}$ **(11)** $30x^4-6x^2+12x$

(12) $\dfrac{5\sqrt{x^3}}{2}$ **(13)** $\dfrac{12-4x^2}{(x^2+3)^2}$ **(14)** $-16x(3-4x^2)$

(15) $\dfrac{2x-3}{(3x-x^2)^2}$ **(16)** $\dfrac{4\sqrt[3]{x}}{3}$ **(17)** $5x^4-4x^3+2x-1$

(18) $\dfrac{4x}{(2-3x)^2}$ **(19)** $x(x^2+3)^4(-13x^3-9x+30)$

(20) $\dfrac{-6x^2-6}{(x^3+3x)^2}$ **(21)** $\dfrac{5x\sqrt{x}}{8}$ **(22)** $4(3x^2+3)(x^3+3x+4)^3$

(23) $\dfrac{8-6x^2}{(x^3-4x)^2}$ **(24)** $(x^3-1)(8x^4+21x^3-2x-3)$

(25) $\dfrac{1-x^2}{(x-1)^4}=\dfrac{-x-1}{(x-1)^3}$ **(26)** $\dfrac{2-4x}{(x+1)^4}$ **(27)** $\dfrac{-1}{18}, 18$

(28) $\dfrac{-4}{25}, \dfrac{25}{4}$ **(29)** $2x-y-4=0$ **(30)** 108

12–increasing and decreasing functions

(1) *A* $y'>0$ increasing *B* $y'>0$ increasing
C $y'=0$ stationary *D* $y'<0$ decreasing
E $y'=0$ stationary *F* $y'>0$ increasing **(2)** *A* $y'<0$
B $y'<0$ *C* $y'=0$ *D* $y'>0$ *E* $y'>0$ *F* $y'=0$ *G* $y'<0$
(3) (a) B and C (b) A and B, C and D
(5) **(i)** stationary **(ii)** increasing **(iii)** increasing
(iv) decreasing **(7)** $x>1$ **(9)** $x<4$ **(10)** **(i)** $x=-1,\ x=3$
(ii) $-1<x<3$ **(11)** **(i)** $x=\dfrac{-1}{3}, 3$ **(ii)** $x<\dfrac{-1}{3},\ x>3$
(12) **(i)** $1\dfrac{1}{2}<x<4$ **(ii)** $x<1\dfrac{1}{2},\ x>4$ **(13)** $x=4, -2$
(14) **(i)** $x=\dfrac{-1}{3}, 4$ **(ii)** $\dfrac{-1}{3}<x<4$ **(iii)** $x<\dfrac{-1}{3},\ x>3$
(15) $x=-4$ (decreasing), $x=-1$ (increasing), $x=0$ (increasing)

13–local (relative) maxima and minima and horizontal points of inflexion using the first derivative

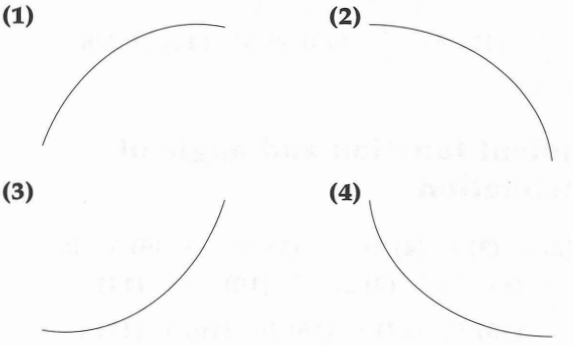
(1) max **(2)** inflexion **(3)** max **(4)** inflexion **(5)** min
(6) inflexion **(7)** inflexion **(8)** min **(9)** min
(10) inflexion **(15)** max, $(0,-6)$ **(16)** min, $(-4,-11)$
(17) max, $\left(\dfrac{1}{2}, 4\right)$ **(18)** max, $(-1, 4)$; min, $(1,-4)$
(19) max, $(-4,28)$; min, $(-1,1)$ **(20)** inflexion **(21)** $\left(\dfrac{3}{2}, 0\right)$
(22) max, $\left(-1, 5\dfrac{2}{3}\right)$; min, $(3,-5)$ **(23)** max, $(2,9)$
(24) min, $(-1,-7)$ **(25)** max, $(-1,2)$; min, $(1,-2)$
(26) max, $(-3,75)$; min, $(0, 48)$

14–the second deriviative

(1) 8 **(2)** 12 **(3)** $24x$ **(4)** $18x$ **(5)** $2+2x^{-3}$ **(6)** $\dfrac{6}{x^4}$
(7) $-24x$ **(8)** $-30x$ **(9)** $20x^3-36x^2$ **(10)** $-42x+60x^2-30x^4$
(11) $\dfrac{3}{x^3}$ **(12)** $\dfrac{2}{x^3}$ **(13)** $\dfrac{-1}{4\sqrt{x^3}}$ **(14)** $\dfrac{-2}{9\sqrt[3]{x^5}}$ **(15)** $384(8x+1)$
(16) $320(3-4x)^3$ **(17)** $\dfrac{-9}{4}(3x+1)^{-\frac{3}{2}}$ **(18)** $\dfrac{-1}{4}(3-x)^{-\frac{3}{2}}$
(19) $\dfrac{2}{(x-1)^3}$ **(20)** $\dfrac{12}{(2-x)^3}$ **(21)** **(i)** 26 **(ii)** -10 **(iii)** -2
(22) **(i)** -2 **(ii)** 16 **(iii)** 32 **(23)** **(i)** 12 **(ii)** -18 **(iii)** -15

15–concavity and the second derivative

(1) **(2)**



(3) **(4)**

(5) *A* $y'>0,\ y''<0$ *B* $y'>0,\ y''<0$
C $y'<0,\ y''<0$ *D* $y'<0,\ y''>0$ *E* $y'>0,\ y''>0$
(6) decreasing, concave up **(7)** increasing, concave up
(8) increasing, concave up **(9)** decreasing, concave down
(10) increasing, concave down **(11)** decreasing, concave down **(12)** decreasing, concave up **(13)** decreasing, concave up **(14)** all reals **(16)** $x<2$ **(17)** $x>2$

16–maxima, minima and points of inflexion using the second derivative

(1) **(i)** *C, F, J* **(ii)** *A, D, H, I* **(iii)** *B, E, G* **(2)** $\left(1, -4\dfrac{2}{3}\right)$
(3) $\left(\dfrac{2}{3}, 19\dfrac{2}{9}\right)$ **(4)** $(2,-55), (-2,-103)$ **(5)** $\left(\dfrac{1}{2}, -4\dfrac{1}{4}\right)$
(8) max, $(-2,13)$; min, $\left(\dfrac{2}{3}, 3\dfrac{14}{27}\right)$; inflexion, $x=\dfrac{-2}{3}$
(9) No pt. of inflexion **(10)** min, $(-4,-18)$ **(11)** max, $(-2, 7)$ **(12)** min, $(0,-6)$ **(13)** max, $(0, 4)$ **(14)** horizontal inflexion, $(0,-7)$ **(15)** max, $(-2, 21)$; min, $(2, -11)$; inflexion, $(0,5)$ **(16)** max, $(-1,8)$; min, $(3,-24)$; inflexion, $(1,-8)$ **(17)** max, $(0,0)$; min, $(5,-125)$; inflexion, $\left(2\dfrac{1}{2}, 62\dfrac{1}{2}\right)$

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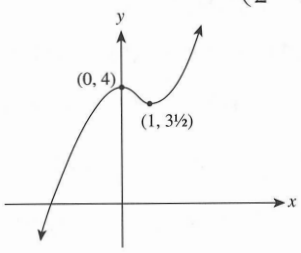
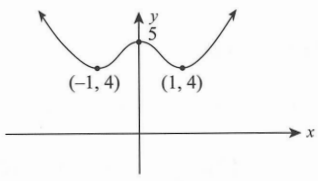
(18) min, $(1,-3)$ **(19)** max, $\left(1, 9\frac{1}{2}\right)$; min, $(0,9)$; inflexion, $\left(\frac{1}{2}, 9\frac{1}{4}\right)$

17–curve sketching using first and second derivative

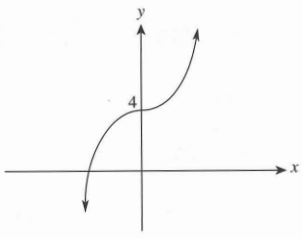
(1) max,$(0,4)$; min,$(1,3\frac{1}{2})$; inflexion, $\left(\frac{1}{2}, 3\frac{3}{4}\right)$; y-intercept 4



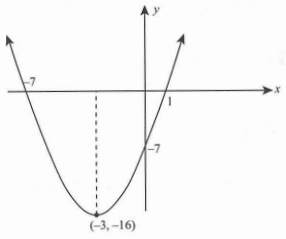
(2) max, $(0,5)$; min, $(1,4)$; min, $(-1,4)$; y-intercept 5



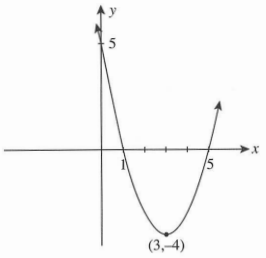
(3) inflexion, $(0,4)$; y-intercept 4



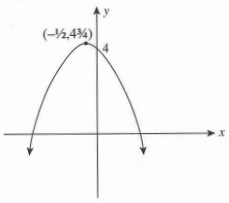
(4) min, $(2,1)$; y-intercept -7



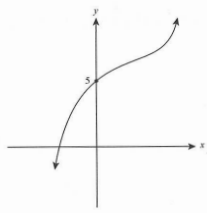
(5) min, $(3,-4)$; y-intercept 5



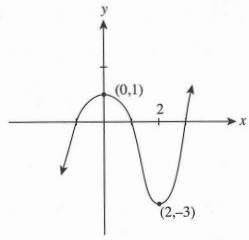
(6) max, $\left(\frac{-1}{2}, 4\frac{3}{4}\right)$; y-intercept 4



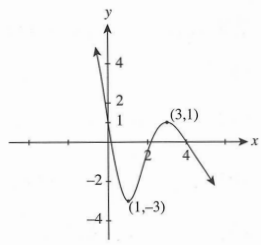
(7) inflexion, $(1,6)$; y-intercept 5



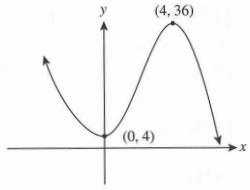
(8) max, $(0,1)$; min, $(2,-3)$; inflexion, $(1,-1)$; y-intercept 1



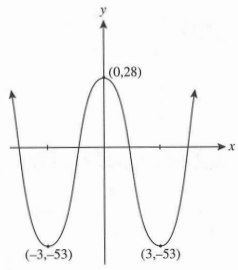
(9) min, $(1,-3)$; max, $(3,1)$; inflexion, $(2,-1)$; y-intercept 1



(10) max, $(4,36)$; min, $(0,6)$; inflexion, $(2,20)$; y-intercept 4



(11) max, $(0,28)$; min, $(-3,-53)$; min, $(3,-53)$; inflexion, $(\sqrt{3},-17)$ and $(-\sqrt{3},-17)$



18–calculus review 3

(1) $x>-2$ **(3)** $x<-4, x>3$ **(4)** max, $x=2$, min, $x=-1$

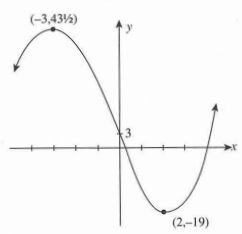
(5) (iii) or (iv) **(6)** 2 **(7)** -2 **(8)** $8-36x+60x^2$

(9) $12x^2+2x-\frac{1}{2}$ **(10)** $192(4x-3)^2$ **(11)** $\frac{-1}{(2x-1)^{\frac{3}{2}}}$

(12) $\frac{8}{(2x+1)^3}$ **(13)** $40x^3+6x+4$ **(14)** $84x^5+40x^3-6x$

(15) concave down **(16)** (i) $x<2$ (ii) $x>2$

(17) $(3,-34)1$ **(18)** min, $(-1,-2)$ **(19)** max, $\left(-3, 43\frac{1}{2}\right)$ min, $(2,-19)$; inflexion, $x=-\frac{1}{2}$



(20) $\frac{dp}{dt}>0, \frac{d^2p}{dt^2}<0$ **(21)** $\frac{dN}{dt}>0, \frac{d^2N}{dt^2}<0$

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19–finding maxima and minima

(1) max 20, min 4 **(2)** max 7, min –9 **(3)** max 24, min –1 **(4)** max $-1\frac{3}{4}$, min –22 **(5)** max 5, min –45, **(6)** max 24, min 19 **(7)** max 1, min –19 **(8)** max 67, min –14 **(9)** max 125, min 1 **(10)** max $4\frac{2}{3}$, min $3\frac{1}{6}$

20–maxima and minima problems

(1) 36 m² **(2)** 1250 m² **(3)** 16,16 **(4)** 128, \$12 303
(5) 100, 50 **(6)** **(a)** $400-\pi r^2$ **(b)** $\frac{400-\pi r^2}{2\pi r}$ **(c)** $\frac{200}{\pi r}-\frac{r}{2}$
(d) $200r-\frac{\pi r^3}{2}$ **(e)** $200-\frac{3\pi r^2}{2}$ **(f)** $-3\pi r$ **(g)** $\sqrt{\frac{400}{3\pi}}$
(7) 5.64 cm, 1128.38 cm² **(8)** 30 cm, 30 cm, 15 cm
(9) 14 m, 7 m

21–primitive function

(1) x^2+c **(2)** x^2+8x+c **(3)** $x^4+\frac{1}{2}x^2+x+c$
(4) x^3+4x^2-5x+c **(5)** $-\frac{3}{4}x^4-x^3+7x+c$
(6) $\frac{1}{4}x^4-\frac{1}{5}x^5+c$ **(7)** $\frac{1}{5}x^5+\frac{1}{7}x^7+c$ **(8)** $\frac{2}{3}x^{\frac{3}{2}}+c$
(9) $-x^{-1}+c$ **(10)** $-\frac{1}{2}x^{-2}+c$ **(11)** $-x^{-2}+x^2+3x+c$
(12) $\frac{3}{2}x^{\frac{3}{2}}+x^2-4x+c$ **(13)** $3x^2+x^4+c$
(14) $x^3+\frac{1}{2}x^4+c$ **(15)** $x^4+\frac{7}{2}x^2-\frac{1}{2}x+c$
(16) $3x^3-2x^2+x-23$ **(17)** $\frac{1}{4}x^4+\frac{1}{2}x^2+4$
(18) $x^3+\frac{7}{2}x^2-x-12$ **(19)** $2x^4-x^3-\frac{1}{2}x^2+14\frac{1}{2}$
(20) $\frac{4}{3}x^3-\frac{1}{4}x^4+x-8\frac{3}{2}$ **(21)** x^2+4x
(22) $\frac{5}{2}x^2-3x-3$ **(23)** x^3-3x^2+3x+2

22–indefinite integrals

(1) $\frac{1}{2}x^2+x+c$ **(2)** $x^3-\frac{1}{2}x^2+c$ **(3)** $x+c$
(4) $x^4+x^3-x^2+c$ **(5)** $\frac{1}{3}x^3-x^4+c$
(6) $\frac{1}{4}x^4-\frac{1}{3}x^3+x^2-5x+c$ **(7)** $\frac{1}{2}x^4-\frac{4}{5}x^5+\frac{1}{2}x^2+c$
(8) $-x^{-1}+\frac{1}{2}x^{-2}+2x+c$ **(9)** $-x^{-2}-2x^{-1}+c$
(10) $x^3-2x^{-2}-\frac{1}{2}x^2+c$ **(11)** $\frac{2}{3}x^{\frac{3}{2}}+\frac{3}{4}x^{\frac{4}{3}}+c$
(12) $\frac{4}{3}x^{\frac{3}{2}}-\frac{3}{5}x^{\frac{5}{3}}+c$ **(13)** $-3x^{-1}+6x^{\frac{2}{3}}+c$
(14) $-\frac{2}{x}+\frac{2}{3x^3}-3x+c$ **(15)** $\frac{2\sqrt{x^3}}{3}-\frac{3\sqrt[3]{x^4}}{4}-x+c$
(16) $8\sqrt{x}-2x+c$ **(17)** $\frac{9\sqrt[3]{x^2}}{2}-\frac{5\sqrt[5]{x^4}}{2}+c$
(18) $\frac{3\sqrt[3]{x^5}}{5}+\frac{2\sqrt{x^5}}{5}+c$ **(19)** $\frac{1}{2}x^2+x+c$
(20) $x^3-\frac{1}{2}x^2+x+c$ **(21)** $-\frac{4}{x}-6x-\frac{3}{2x^2}+c$
(22) $(x+3)^3+c$ **(23)** $(x-2)^5+c$ **(24)** $(2x-1)^3+c$
(25) $\frac{1}{3}(x-5)^3+c$ **(26)** $-\frac{1}{4}(4-x)^4+c$
(27) $-\frac{1}{12}(4-3x)^4+c$ **(28)** $\frac{1}{10}(4+2x)^5+c$
(29) $\frac{2}{3}\sqrt{(x+7)^3}+c$ **(30)** $\frac{1}{3}\sqrt{(2x-1)^3}+c$
(31) $\frac{3}{4}\sqrt[3]{(x-1)^4}+c$ **(32)** $-\frac{1}{4}\sqrt[3]{(9-3x)^4}+c$
(33) $\sqrt{(2x-5)}+c$ **(34)** $\frac{1}{2}\sqrt{4x-5}+c$
(35) $-\frac{2}{5}\sqrt{6-5x}+c$ **(36)** $4\sqrt{2x-8}+c$

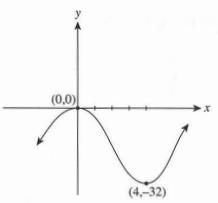
23–definite integrals

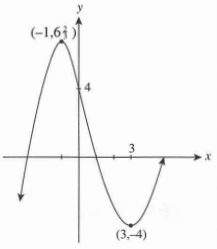
(1) 8 **(2)** 65 **(3)** 18 **(4)** 3 **(5)** 2 **(6)** $10\frac{2}{3}$
(7) $8\frac{2}{3}$ **(8)** $-11\frac{1}{3}$ **(9)** $2\frac{2}{3}$ **(10)** –1 **(11)** $\frac{1}{8}$
(12) $3\frac{1}{2}$ **(13)** $-\frac{5}{8}$ **(14)** $-10\frac{3}{4}$ **(15)** $1\frac{2}{3}$
(16) $\frac{3}{8}$ **(17)** $-\frac{1}{4}$ **(18)** $12\frac{2}{3}$ **(19)** **(i)** $7\frac{5}{6}$ **(ii)** $\frac{5}{6}$
(iii) $8\frac{2}{3}$ **(iv)** $8\frac{2}{3}$ **(20)** $10\frac{2}{3}$ **(21)** $-29\frac{1}{4}$

24–calculus review 4

(1) x^6+x^2+c **(2)** x^3+2x^2+x+c **(3)** $x^5+\frac{1}{2}x+c$
(4) $\frac{x^4}{4}+2x^2-5x+c$ **(5)** $\frac{x^5}{5}-x^2+3x+c$ **(6)** $x+c$
(7) $\frac{x^2}{2}-x^3+c$ **(8)** $\frac{4}{3}\sqrt{x^3}+c$ **(9)** $-\frac{1}{x^2}-\frac{1}{x}+c$
(10) $\frac{3}{2}x^{\frac{2}{3}}-\frac{1}{2}x^2-4x+c$ **(11)** $x^3-\frac{9}{2}x^2-12x+c$
(12) $\frac{2}{3}\sqrt{(x+1)^3}+c$ **(13)** $y=\frac{3}{2}x^2-4x+5$
(14) $y=3x^4-2x^2+6$ **(15)** x^2-x+c
(16) $\frac{1}{4}x^4+3x^2-5x+c$ **(17)** $x^5-\frac{1}{4}x^4-4x+c$
(18) $x^5-\frac{3}{2}x^2+5x+c$ **(19)** $\frac{1}{2}x^2-\frac{4}{3}x^3+3x+c$
(20) $3x^3-16x+c$ **(21)** $x^4-x^3+\frac{1}{2}x^2+c$
(22) $\frac{1}{3}(x-4)^3+c$ **(23)** $\frac{2}{3}\sqrt{x^3}-\frac{3}{2}x^2+c$
(24) $\frac{2}{3}\sqrt{x}+4x+c$ **(25)** 16 **(26)** –1 **(27)** 12 **(28)** $-3\frac{1}{3}$
(29) $31\frac{1}{3}$ **(30)** $-\frac{1}{8}$
(31) 26 **(32)** 60

(33) max, (0, 0);
min, (4, –32);
inflexion, (2, –16)





(34) max, $\left(-1,\ 6\frac{2}{3}\right)$, min, (3, –4)
(35) 8, \$28750 **(36)** 108 m²

25–area between a curve and the *x*-axis

(1) $2\frac{1}{3}$ *units*² **(2)** 29 *units*² **(3)** 4 *units*² **(4)** $\frac{2}{3}$ *units*²
(5) 4 *units*² **(6)** $\frac{5}{6}$ *units*² **(7)** 8 *units*² **(8)** $12\frac{2}{3}$ *units*²

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(9) $16\,units^2$ **(10)** $1\frac{1}{3}\,units^2$ **(11)** $\frac{1}{6}\,units^2$

(12) $57\frac{1}{6}\,units^2$ **(13)** $4\frac{1}{2}\,units^2$ **(14)** $4\frac{1}{2}\,units^2$

(15) $3\frac{1}{12}\,units^2$ **(16)** $8\,units^2$

26–area between a curve and the *y*-axis and the area between two curves

(1) $18\,units^2$ **(2)** $11\frac{1}{4}\,units^2$ **(3)** $22\frac{2}{3}\,units^2$

(4) $23\frac{1}{3}\,units^2$ **(5)** $\frac{19}{27}\,units^2$ **(6)** $\frac{3}{2}\,units^2$

(7) $2\,units^2$ **(8)** (i) 0 (ii) 2 (iii) –2 $4\,units^2$ **(9)** $4\frac{1}{2}\,units^2$

(10) $40\frac{1}{2}\,units^2$ **(11)** $\frac{3}{10}\,units^2$ **(12)** $9\frac{3}{5}\,units^2$

(13) $4\frac{1}{2}\,units^2$ **(14)** $21\frac{1}{3}\,units^2$ **(15)** $72\,units^2$

(16) $10\frac{2}{3}\,units^2$ **(17)** between curve & *y*-axis by $\frac{1}{3}\,units^2$

27–simpson's rule and the trapezoidal rule

(1) $2\frac{1}{3}\,units^2$ **(2)** $3\frac{1}{3}\,units^2$ **(3)** $20\,units^2$ **(4)** 68

(5) –80 **(6)** $-5\frac{1}{3}$ **(7)** $1\frac{1}{3}\,units^2$ **(8)** $14\,units^2$

(9) 8. **(10)** $\frac{17}{24}$ **(11)** $12\frac{1}{2}\%$ **(12)** $25.1\,units^2$

(13) 26.53 m^2

28–derivatives of logarithmic and exponential functions

(1) $6e^{6x}$ **(2)** $4e^{4x-1}$ **(3)** $-3e^{-3x}$ **(4)** $\frac{1}{4}e^x$ **(5)** $6x^2e^{x^3}$

(6) $1+e^{4x}$ **(7)** $-2e^{2x-3}$ **(8)** $-2xe^{x^2+1}$ **(9)** $2+5e^{4-5x}$

(10) $2e^{2x}+3e^{3x}$ **(11)** $-2xe^{-x^2}+e^{-x}$ **(12)** $1+3e^{3x-4}$

(13) $\frac{2}{x}$ **(14)** $\frac{1}{x}$ **(15)** $\frac{2}{2x-5}$ **(16)** $\frac{8x}{4x^2-1}$

(17) $\frac{6}{2x-1}$ **(18)** $\frac{2}{6x-1}$ **(19)** $\frac{1}{x+2}-\frac{1}{x-1}$

(20) $\frac{2}{2x+3}-\frac{2}{2x-1}$ **(21)** $\frac{4}{4x-3}-\frac{2}{2x+1}$

(22) $\frac{1}{x+3}+\frac{1}{x-2}$ **(23)** $\frac{1}{x-3}+\frac{1}{x+4}$ **(24)** $\frac{2}{2x-1}+\frac{1}{x+2}$

(25) $\frac{2}{x+3}$ **(26)** $\frac{6}{2x-1}$ **(27)** $\frac{-8}{3-4x}$ **(28)** $\frac{4}{2x-1}+\frac{3}{3x-2}$

(29) $\frac{1}{2(x-1)}$ **(30)** $\frac{1}{2x+1}$ **(31)** $\frac{-1}{1-3x}$ **(32)** $\frac{-1}{x}$

(33) $\frac{-2}{2x-5}$ **(34)** $\frac{-3}{3x-4}$ **(35)** $\frac{-1}{2(x-3)}$ **(36)** $\frac{1}{7-2x}$

(37) $\frac{-1}{3x+1}$

29–integration of logarithmic and exponential functions

(1) e^x+c **(2)** $e^{3x}+c$ **(3)** $e^{1-4x}+c$ **(4)** $\frac{1}{2}e^{2x}+c$

(5) $\frac{1}{4}e^{4x+1}+c$ **(6)** $\frac{-1}{2}e^{3-2x}+c$ **(7)** $e^{x^2+1}+c$

(8) $e^{x^3+1}+c$ **(9)** $\frac{1}{2}e^{x^2-5}+c$ **(10)** $\frac{-1}{6}e^{-3x^2}+c$

(11) $\log_e(2x+1)+c$ **(12)** $\log_e(7x+4)+c$

(13) $\frac{1}{3}\log_e(3x+4)+c$ **(14)** $\frac{1}{9}\log_e x+c$

(15) $\frac{1}{5}\log_e(5x-1)+c$ **(16)** $\frac{-1}{9}\log_e(2-9x)+c$

(17) $\log_e(x^2-1)+c$ **(18)** $\log_e(3x^2-4)+c$

(19) $\frac{1}{2}\log_e(x^2+2)+c$ **(20)** $\frac{3}{4}\log_e(4x-1)+c$

(21) $\frac{-1}{6}\log_e(2-3x^2)+c$ **(22)** $\frac{-3}{8}\log_e(1-4x^2)+c$

(23) $x^2+4x-\log_e x+c$ **(24)** $\frac{1}{2}x^2-7x-3\log_e x+c$

(25) $x-4\log_e(x+2)+c$ **(26)** $x-3\log_e(x+2)+c$

(27) $2x+\log_e(x-7)+c$ **(28)** $x+\frac{1}{2}\log_e(2x-1)+c$

(29) $x-3\log_e(x-1)+c$ **(30)** $x+\frac{3}{2}\log_e(2x+1)+c$

30–further logarithmic and exponential functions

(1) $3e^x$ **(2)** $2e^{2x+1}(2x+1)$ **(3)** $2xe^{x^2}(x^2+1)$

(4) $\frac{2x}{x+1}+2\ln(x+1)$ **(5)** $\frac{2x^3}{x^2+1}+2x\ln(x^2+1)$ **(6)** $\frac{1-x}{e^x}$

(7) $e^{x+1}\left(\frac{1}{x}+\ln x\right)$ **(8)** $e^{2x}\left(\frac{1}{x+3}+2\ln(x+3)\right)$ **(9)** $\frac{2(1-2x)}{e^{2x}}$

(10) $\frac{1}{e^x}\left(\frac{1}{x}-\ln x\right)$ **(11)** 1 **(12)** $\frac{-2}{3}$ **(13)** $4e^4$ **(14)** –2

(15) 0.69 **(16)** 0.17 **(17)** 0.46 **(18)** 0.23 **(19)** 4.67

(20) 38.17 **(21)** 1.43 **(22)** 4.92 **(23)** $2e(e^2-1)$

(24) $\frac{-1}{3}(1-e^6)$ **(25)** $\frac{1}{3}e^3-\frac{1}{2}e^2+\frac{1}{6}$ **(26)** ln 2 **(27)** 0.51

(28) 6 + ln 2

31–differentiation of trigonometric functions

(1) $3\cos x$ **(2)** $-4\sin x$ **(3)** $9\sec^2 x$ **(4)** $-3\sin x-2\sec^2 x$

(5) $\frac{1}{2}(2\cos x-5\sin x)$ **(6)** $\frac{1}{2}\cos x+\frac{1}{4}\sin x-6\sec^2 x$

(7) $-4\sin 4x$ **(8)** $6\sec^2 6x$ **(9)** $\frac{2}{3}\cos\frac{2}{3}x$ **(10)** $12\sec^2 3x$

(11) $-\sin 3x$ **(12)** $\sin x\cos x$ **(13)** $-4\cos 2x\sin 2x$

(14) $6\tan 3x\sec^2 3x$ **(15)** $-\tan x$ **(16)** $\cot x$

(17) $x(x\cos x+2\sin x)$ **(18)** $3x^2(\cos 3x-x\sin 3x)$

(19) $4x(2x\sec^2 4x+\tan 4x)$ **(20)** $e^{2x}(\cos x+2\sin x)$

(21) $\sec x\tan x$ **(22)** $-\cot 2x\ \operatorname{cosec} 2x$

(23) $\sec x(1+x\tan x)$

(24) $\frac{2x\sin x-x^2\cos x}{\sin^2 x}=x\operatorname{cosec} x[2-x\cot x]$ **(25)** $2\sec^2 x$

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(26) $\cos 2x - 2x\sin 2x$ **(27)** $\frac{1}{\sqrt{2}}$ **(28)** $-\frac{1}{2}$ **(29)** 2

(30) $\frac{3}{2} - 2\sqrt{3}$ **(31)** max at $\sqrt{2}$, min at $-\sqrt{2}$

(32) max at 2, min at -2

32–integration of trigonometric functions

(1) $-4\cos x + c$ **(2)** $2\sin x + c$ **(3)** $3\tan x + c$

(4) $\frac{-1}{3}\cos 3x + c$ **(5)** $\frac{1}{4}\sin 4x + c$ **(6)** $\frac{3}{5}\cos 5x + c$

(7) $\frac{-5}{4}\cos 4x + c$ **(8)** $\frac{3}{2}\sin 2x + c$ **(9)** $\frac{1}{2}\tan 4x + c$

(10) $\frac{3}{2}\sin 4x + c$ **(11)** $-4\cos\frac{x}{4} + c$ **(12)** $-3\sin\frac{x}{3} + c$

(13) $-\cos\frac{x}{3} + c$ **(14)** $\frac{-2}{15}\sin 5x + c$ **(15)** $\frac{1}{2}\sin(2x-4) + c$

(16) $\frac{1}{3}\cos(1-3x) + c$ **(17)** $\frac{1}{4}\sin 2x - \frac{1}{2}\cos 2x + c$

(18) $2\sin 2x - \frac{3}{4}\tan 4x + c$ **(19)** $\tan x - x + c$ **(20)** 1

(21) 0 **(22)** $\frac{1}{\sqrt{3}}$ **(23)** $1 - \frac{1}{\sqrt{2}}$ **(24)** $-\sqrt{2} - 3$ **(25)** $\frac{\pi}{2} - 3$

(26) $\frac{1}{2}$ **(27)** $2 - \frac{2}{\sqrt{3}}$ **(28)** $\pi - 1$ **(29)** $y = 4 - 3\cos x$

(30) $y = \frac{1}{2}\sin x - \frac{1}{2\sqrt{2}}$ **(31)** 2 *units*^{2} **(32)** $\left(1 - \frac{1}{\sqrt{2}}\right)$ *units*^{2}

(33) $(3 - \sqrt{3})$ *units*^{2} **(34)** $\left(\frac{\pi}{4} + \frac{1}{\sqrt{2}}\right)$ *units*^{2}

33–volumes of solids of revolution

(1) $\frac{32\pi}{5}$ *units*^{3} **(2)** $\frac{127\pi}{7}$ *units*^{3} **(3)** $\frac{206}{15}\pi$ *units*^{3}

(4) cone, 9π *units*^{3} **(5)** $\frac{\pi}{30}$ *units*^{3} **(6)** $\frac{32}{3}\pi$ *units*^{3}

(7) $\frac{4\pi}{3}$ *units*^{3} **(8)** 2π *units*^{3} **(9)** 8π *units*^{3} **(10)** 2π *units*^{3}

(11) $\frac{3\pi}{4}$ *units*^{3} **(12)** $\frac{\pi}{2}(e^6 - e^2)$ *units*^{3} **(13)** $\frac{2\pi}{\sqrt{3}}$ *units*^{3}

(14) $\frac{72\pi}{5}$ *units*^{3} **(15)** $\frac{64\pi}{15}$ *units*^{3}

34–displacement, velocity and acceleration

(1) (i) 3 m to right of O

(ii) 35 m to right of O, 28 m/s, 18 m/s^2

(2) 23 m to left of O, 12 m/s^2 **(3) (i)** t = 3 sec **(ii)** 54 m

(iii) –54 m/s **(4) (i)** 5 m to right of O **(ii)** $7\frac{1}{2}$ s **(iii)** 5 m

(5) (i) 7 m **(ii)** 16 m **(iii)** 7 sec, -8 m/s

(6) 5 m to left of O **(7) (i)** $\frac{\pi}{4}$ sec, $\frac{3\pi}{4}$ sec **(ii)** $(1+\sqrt{2})$ m/s

(iii) $xt + \sqrt{2}\cos t - \sqrt{2}$

35–calculus review 5

(1) $6e^{6x}$ **(2)** $24xe^{3x^2+5}$ **(3)** $\frac{3}{3x-5}$ **(4)** $\frac{14x}{7x^2-2}$

(5) $\frac{1}{x+3} + \frac{2x+4}{x^2+4x}$ **(6)** $\frac{6}{6x-1} - \frac{2}{2x+4}$

(7) $\frac{3}{3x+4} - \frac{2}{2x-1}$ **(8)** $\frac{6}{2x-1} + 3\ln(2x-1)$ **(9)** $\frac{5-15x}{e^{3x}}$

(10) $4\cos 4x$ **(11)** $-2x\sin(x^2-3)$ **(12)** $-2\sin 3x$

(13) $2\sec^2 4x$ **(14)** $3\sin^2 x\cos x$ **(15)** $\cos x - x\sin x$

(16) $e^x(3\sec^2 3x + \tan 3x)$ **(17)** $2e^{2x} + c$

(18) $\frac{3}{2}\ln(2x-1) + c$ **(19)** $\ln(1+x^2) + c$ **(20)** $\frac{1}{2}e^{2x^2+1} + c$

(21) $5\sin x$ **(22)** $-4\cos x + c$ **(23)** $\frac{-3}{2}\cos 2x + c$

(24) $\frac{1}{2}\sin 2x + c$ **(25)** $\frac{1}{3}\tan 3x + c$ **(26)** $\frac{3}{4}\tan 4x + c$

(27) 0.5 **(28)** $\frac{1}{4}$ **(29)** 0.55 **(30)** $x + 7\ln(x-3) + c$

(31) $x - 2\ln(2x+3) + c$ **(32)** 63 *units*^{2} **(33)** $2\frac{1}{6}$ *units*^{2}

(34) $4\frac{2}{3}$ *units*^{2} **(35)** $\frac{25}{36}$ *units*^{2} **(36)** 259.2π *units*^{3}

(37) $\pi(\sqrt{5} - 1)$ *units*^{3}

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